Finite Spectral Shadows for the Collatz Valuation Cocycle

John A. Janik
john.janik@gmail.com

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Abstract

We study residue equidistribution for the accelerated Collatz/Syracuse map $S(x) = (3x + 1)/2^{v_2(3x+1)}$ through a pathwise twisted Birkhoff functional—the *deterministic defect cocycle* $W_n(x; \chi, s)$—and the finite residue–height transfer operators $\mathcal{L}_{s,Q,L,\chi}$ that govern it. Our aim is not to prove the Collatz conjecture but to identify, rigorously, the correct finite spectral object and the correct quotient on which its gap should be sought. We prove two clean finite identities: a *sliding-window congruence* expressing $x_n \bmod 3^r$ as a finite window of the valuation word, and the fact that the quadratic character χ_2 of $(\mathbb{Z}/3^r)^\times$ satisfies $\chi_2(S(x)) = (-1)^{v_2(3x+1)}$, i.e. χ_2 is *valuation parity in disguise* and is an exact coboundary of the faithful residue–valuation skew product. Consequently the right form of the finite spectral hypothesis ("H23a") is a gap for all *non-coboundary* characters of the *faithful* (synchronized) operator, not for all nontrivial characters. We then prove three theorems about that operator: a *classification of coboundaries* ($\mathrm{Cob} = \langle \chi_2 \rangle$, for arbitrary positive branch weights), whence by Wielandt's theorem a *strict* spectral gap for every non-coboundary character at every finite level and tilt; a *conductor collapse* (the nonzero spectrum of a conductor-3^c sector is computed at level 3^c), which reduces uniformity of the gap in Q to the boundedness of a single sequence of computable constants $\gamma_c(s)$; and a *rotation identity* showing the quadratic sector of the killed height-strip operator is a rigid rotation of the principal one. The uniformity question is then *resolved*: a threshold $(m = c)$ Schur analysis of the adjoint kernel yields bounded Plancherel shells for the renewal constants ($\tilde{Q}_L \le 3C_1 < 5.16$), cascade row contraction at rate $(\sqrt{3}/2)^c$, a level-cost law with the exact constant $\frac{4}{7} < 3^{-1/2}$, and a root-ball analysis closing the degenerate budget—together the finite H23a theorem $\sup_c \gamma_c(0) \le (\sqrt{3}/2)^{1/2} + o(1) < 1$ at $s = 0$. On the global side we prove the *keystone arrow* (every bad orbit casts a finite non-coboundary two-point shadow), reduce the remaining quenched exclusion to a joint $(2, 3)$-adic max-plus rigidity, and prove two unconditional exclusion theorems: periodic traps via linear forms in logarithms, and—by a new *repetition rigidity* argument in which repeated valuation blocks are exact cycles—all bounded-discrepancy valuation words of subexponential complexity, in particular every Sturmian word. A counterexample must therefore mimic randomness at every banded scale. The elementary core (the 2-adic coding lemma and repetition rigidity) is machine-verified in Lean 4. A companion computational report [16] documents the numerical suite.

Contents

Convention. From part II onward the residue space is specialised to $U_{3^r} = (\mathbb{Z}/3^r\mathbb{Z})^\times$, $Q = 3^r$, and all characters are multiplicative unless stated otherwise. The additive-character experiments reported in the companion are diagnostic only and are not the final operator vehicle. The character twist in the transfer operator is always placed on the *source* vertex a (see part II); with endpoint twisting the coboundary formulae are conjugate but take a different form.

Part I

The object and the open inverse principle

1 The obstacle that fixes the shape of the object

A single odd integer x_0 produces one deterministic valuation word $w(x_0) = (b_0, b_1, \dots)$, $b_i = v_2(3x_i + 1) \geq 1$, with $x_{i+1} = S(x_i)$. (For background on the $3x+1$ problem and its stochastic models see the survey and collection of Lagarias [20, 21] and the annealed analyses of Sinai, Kontorovich–Lagarias, and Tao [33, 19, 36].) The annealed law is the pushforward of Haar measure on $\mathbb{Z}_2$ under $x \mapsto v_2(3x+1)$, i.e. $\Pr(b) = 2^{-b}$. "The orbit obeys the annealed law" asserts that the one prescribed word $w(x_0)$ equidistributes for the residue–height skew product. No ergodic theorem applies to a single prescribed non-generic point, so the missing object cannot be a measure-theoretic average.

It must instead be a *dichotomy with a closure*: a pathwise quantity forced either to decay (the orbit equidistributes) or to localise as a finite, excludable obstruction. It must reduce, at the trivial character and after ensemble averaging, to the annealed pressure and the Cramér tilt; and its localisation must land in a fixed finite character sector.

2 The deterministic defect cocycle

For each modulus Q, height truncation L, tilt s, and character χ of the residue space, define the pathwise twisted Birkhoff functional

$$W_n(x;\chi,s) = \frac{1}{n}\sum_{i=0}^{n-1}\chi(x_i \bmod Q)\, 2^{\,s\Delta_{b_i}}, \qquad \Delta_b = \log_2 3 - b.$$

The family $\{W_n(x;\chi,s)\}$ fibered over the finite levels is the *deterministic defect cocycle* $\mathfrak{D}_{\mathrm{Coll}}$. At $s=0$ it is the residue Weyl sum [38], so $x_i \bmod Q$ equidistributes iff $W_n(x;\chi,0)\to 0$ for all $\chi\neq 1$. At the trivial character the ensemble average reproduces the annealed pressure $M(s)=\sum_{b\geq 1}2^{-b}2^{s\Delta_b} = 3^s/(2^{1+s}-1)$ and its Cramér root $M(1)=1$ with tilted law $p_b^{(1)} = 3\cdot 4^{-b}$ (exponential tilting in the sense of [8, 10]; for these quantities in the Collatz setting see [19, 36]). Thus $\mathfrak{D}_{\mathrm{Coll}}$ is the quenched lift of the thermodynamic formalism [29, 3].

At finite level the intended content is, informally, an equivalence between equidistribution failure, an approximate coboundary along the orbit, and a near-peripheral eigenstate of $\mathcal{L}_{s,Q,L,\chi}$. We use only the safe direction.

Remark. The full three-way equivalence is too strong. A non-decaying twisted sum signals correlation with χ but need not be an exact coboundary (it may reflect concentration, subsequential trapping, or nonstationarity); and producing a near-*top* eigenstate is precisely the open keystone. The load-bearing implication is: a persistent W_n bias $\Rightarrow$ a nontrivial finite character shadow, and (conjecturally) a near-peripheral state of $\mathcal{L}_{s,Q,L,\chi}$.

3 The inverse principle (conjectural)

Quenched Inverse Principle (conjectural)

Let x_0 have an infinite forward orbit and suppose, for some *non-coboundary* character χ on some U_{3^r}, some tilt s, and some $\delta > 0$, that $|W_n(x_0;\chi,s)| \geq \delta$ along blocks of length $\ell_j \to \infty$. Then there exist a finite pair (Q,L) and a χ-sector eigenstate of $\mathcal{L}_{s,Q,L,\chi}$ with $|\lambda| \geq \rho(\mathcal{L}_{s,Q,L,0}) - o(1)$, exhibited by the orbit's empirical measure. Conjoined with the non-coboundary gap $\rho(\mathcal{L}_{s,Q,L,\chi}) \leq (1-\kappa_Q)\rho(\mathcal{L}_{s,Q,L,0})$, this is a contradiction. Hence every orbit satisfying the hypotheses has *no persistent non-coboundary finite spectral shadow.*

Remark (Scope of the conclusion). This does not by itself give full equidistribution. It excludes persistent non-coboundary finite spectral shadows. If, in addition, every bounded-deficit or large-deviation failure of the annealed law produces such a shadow—the keystone inverse arrow—then the relevant quenched obstruction is excluded. The keystone is open and is morally a normality/disjointness statement for the 2-adic word, of Sarnak-hard flavour [30].

The two leaks. Closing the principle quantitatively requires (i) a *finiteness* input: a uniform gap κ_Q that does not decay faster than the entropy production of the bias (the subject of part IV); and (ii) a *circularity* input: reading a top eigenfunction off the orbit's own empirical measure presumes near-stationarity, which is what one is proving. The companion report shows the second leak is real at the single-orbit level.

Part II

Rigorous finite identities

These identities are the rigorous core: they bridge the valuation word and the residue characters, and they pin down the unique structural degeneracy of the operator.

Convention. The transfer operator $\mathcal{L}_{s,Q,L,\chi}$ has matrix element, on an edge $(a, z) \xrightarrow{b} (a', z')$ with $a' \equiv (3a + 1)2^{-b}$ and $z' = z + \lfloor R(\log_2 3 - b)\rceil$ (killed off $[0, L]$),

$$\mathcal{L}_{s,Q,L,\chi}[(a', z'), (a, z)] = 2^{-b}2^{s\Delta_b}\,\chi(a),$$

i.e. the character twist sits on the *source* residue a.

4 The sliding-window congruence

Lemma 1 (Sliding-window congruence)**.** *With $B_n = \sum_{i<n} b_i$ and the exact affine iterate $x_n = (3^n x_0 + A_n)/2^{B_n}$, $A_n = \sum_{j=0}^{n-1} 3^{\,n-1-j}2^{B_j}$, one has for $n \geq r$*

$$x_n \equiv \sum_{k=1}^{r} 3^{\,k-1}\,2^{-(b_{n-1}+\cdots+b_{n-k})} \pmod{3^r}.$$

Proof. Modulo 3^r the term $3^n x_0$ vanishes for $n \geq r$. In A_n the summand indexed by j carries $3^{\,n-1-j}$, which is nonzero mod 3^r only when $n - 1 - j < r$, i.e. $j \in \{n - r, \ldots, n - 1\}$. Writing $j = n - k$ ($k = 1, \ldots, r$) gives $3^{\,n-1-j} = 3^{\,k-1}$, so $A_n \equiv \sum_{k=1}^{r} 3^{k-1}2^{B_{n-k}} \pmod{3^r}$. Since 2 is a unit mod 3^r, multiply by 2^{-B_n}: $x_n \equiv \sum_{k=1}^{r} 3^{k-1}2^{B_{n-k}-B_n} = \sum_{k=1}^{r} 3^{k-1}2^{-(b_{n-k}+\cdots+b_{n-1})} \pmod{3^r}$. $\square$

Thus a finite 3^r-residue character is a finite sliding-window observable of the valuation process—one of the few rigorous bridges from valuation word to residue shadow. Since 2 is a primitive root mod 3^r (order $\phi = 2 \cdot 3^{r-1}$), the resonant frequencies attached to nontrivial characters are $\xi = k/(2 \cdot 3^{r-1})$, and the single-symbol Fourier transform $\hat{p}(\xi) = \sum_{b\geq 1} 2^{-b}e(b\xi) = e(\xi)/(2 - e(\xi))$ obeys $|\hat{p}(\xi)| = (5 - 4\cos 2\pi\xi)^{-1/2} < 1$ for $\xi \notin \mathbb{Z}$—the annealed decorrelation mechanism. (The companion confirms the *exact* operator departs from this i.i.d. value.)

5 The quadratic character is valuation parity

Lemma 2 (χ_2 factors through reduction mod 3)**.** *U_{3^r} is cyclic of order $\phi = 2\cdot 3^{r-1}$, and its unique order-2 character χ_2 has $\ker\chi_2 = \{u : u \equiv 1 \pmod 3\}$. Equivalently $\chi_2(a) = +1$ if $a \equiv 1$ and -1 if $a \equiv 2 \pmod 3$.*

Proof. 2 is a primitive root mod 3^r, so the group is cyclic and $\ker\chi_2$ is its unique index-2 subgroup (the squares), of order 3^{r-1}. Reduction $U_{3^r} \to (\mathbb{Z}/3)^\times \cong \mathbb{Z}/2$ has kernel $\{u \equiv 1 \pmod 3\}$, also of order 3^{r-1}; in a cyclic group the subgroup of a given order is unique, so the two coincide. □

Theorem 1 (Key identity)**.** *For every odd x, with $b = v_2(3x+1)$,*

$$\chi_2\big(S(x) \bmod 3^r\big) = (-1)^b.$$

Hence $\chi_2(x_{i+1}) = (-1)^{b_i}$ along every orbit.

Proof. Work mod 3: $2 \equiv -1$ and $3x+1 \equiv 1$, so $S(x) = (3x+1)2^{-b} \equiv (-1)^b \pmod 3$. By lemma 2, $\chi_2(S(x)) = (-1)^b$. (Since $3x+1 \equiv 1 \pmod 3$, $S(x)$ is a unit.) □

Corollary 1. *$\prod_{i=1}^n \chi_2(x_i) = (-1)^{B_n}$: the accumulated quadratic character is exactly the parity of the cumulative 2-adic valuation. In particular the residues are* not *uniform mod 3; they occur in ratio $\Pr(x\equiv 1) = \sum_{b\ even} 2^{-b} = \frac{1}{3}$, $\Pr(x \equiv 2) = \frac{2}{3}$.*

6 The quadratic coboundary and the operator dichotomy

Consider the synchronized residue–valuation skew product with state $(a,\tau) \in U_{3^r} \times (\mathbb{Z}/2)$ and dynamics $(a,\tau) \xrightarrow{b} \big((3a+1)2^{-b},\ \tau + b\big)$ (one valuation b drives both coordinates), and let $\mathcal{L}_\chi$ be the corresponding twisted annealed operator (part II), $\mathcal{L}_{\mathbf{1}}$ the principal one.

Theorem 2 (Quadratic coboundary)**.** *The phase $d(a,\tau) = \chi_2(a)(-1)^\tau \in \{\pm 1\}$ satisfies $d(a',\tau')/d(a,\tau) = \chi_2(a)$ on every edge. Hence $\mathcal{L}_{\chi_2} = D\,\mathcal{L}_{\mathbf{1}}\,D^{-1}$ with $D = \mathrm{diag}(d)$, so $\rho(\mathcal{L}_{\chi_2}) = \rho(\mathcal{L}_{\mathbf{1}})$: the χ_2 sector has* no gap*, for every tilt s and any height discretisation.*

Proof. By theorem 1 $\chi_2(a') = (-1)^b$, and $\tau' = \tau + b$ gives $(-1)^{\tau'} = (-1)^\tau\chi_2(a')$. Thus $d(a',\tau') = \chi_2(a')(-1)^{\tau'} = (-1)^\tau$ and $d(a',\tau')/d(a,\tau) = (-1)^\tau/(\chi_2(a)(-1)^\tau) = \chi_2(a)$. With the source twist, the conjugated matrix element is $d(a',\tau')^{-1}2^{-b}2^{s\Delta_b}\chi_2(a)d(a,\tau) = 2^{-b}2^{s\Delta_b}$, the principal element; D is real orthogonal, so the operators are isospectral. □

So χ_2 is not a defect—it belongs to the height/valuation reference. This forces the operator question.

Proposition 1 (Residue-only gap; the $\mathcal{L}$ dichotomy)**.** *Without the valuation-parity coordinate, χ_2 is not a coboundary: on the $\{\pm 1\}$ (mod 3) reduction the twisted 2×2 operator has eigenvalues 0 and $\sum_b (-1)^b w_s(b)$, giving the closed-form residue-only gap ratio*

$$\frac{\rho(\mathcal{L}^{\mathrm{res}}_{\chi_2})}{\rho(\mathcal{L}^{\mathrm{res}}_{\mathbf{1}})} = \frac{\big|\sum_b (-1)^b w_s(b)\big|}{\sum_b w_s(b)} = \frac{2^{1+s}-1}{2^{1+s}+1} \qquad \Big(\tfrac{1}{3}\ \textit{at } s{=}0,\ \tfrac{3}{5}\ \textit{at } s{=}1\Big), \quad w_s(b) = 2^{-b}2^{s\Delta_b}.$$

Remark (Faithful `sync` vs. `factored`; why faithfulness is forced)*.* Two finite operators couple residue and height. The *faithful* (`sync`) operator uses one valuation b for both, matching the orbit; the `factored` operator $\mathcal{L}_{\mathrm{res}}(\chi) \otimes T_{\mathrm{height}}$ draws the height's valuation independently. The factored operator reports a gap at χ_2, but corollary 1 shows $W_n(x; \chi_2, 0) \to -\frac{1}{3} \neq 0$: the χ_2 component does not decay. The factored operator therefore predicts decay that does not occur. Because the inverse principle runs *backwards* (a persistent W_n must force a peripheral eigenstate), the operator's spectral radius must equal the true growth rate of W_n; only `sync` qualifies. We adopt the faithful `sync` operator and use `factored`/residue-only operators only as diagnostics.

7 Classification of coboundaries and the strict gap at every level

We now prove that χ_2 is the *only* nontrivial coboundary of the synchronized skew product—the central finite theorem of this paper. Throughout this section the operator is the synchronized residue–parity operator of theorem 2: states $(a, \tau) \in U_{3^r} \times (\mathbb{Z}/2)$, edges $(a, \tau) \xrightarrow{b} (S_b(a), \tau + b)$ with $S_b(a) = (3a + 1)2^{-b}$, source twist, and *arbitrary* positive summable branch weights $w_b > 0$ (in particular $w_b = 2^{-b} 2^{s\Delta_b}$ for any tilt s). We allow the twisted-conjugacy (Wielandt [39]) form with a global phase.

Theorem 3 (Coboundary classification)**.** *Let χ be a character of U_{3^r}, and suppose there exist $\lambda \in S^1$ and $d : U_{3^r} \times (\mathbb{Z}/2) \to S^1$ with*

$$d\big(S_b(a), \tau + b\big) = \lambda\, \chi(a)\, d(a, \tau) \qquad \textit{for all } a \in U_{3^r},\ b \geq 1,\ \tau \in \mathbb{Z}/2.$$

Then $\chi \in \{\mathbf{1}, \chi_2\}$. Conversely both occur ($\chi = \mathbf{1}$ with $d \equiv 1$; $\chi = \chi_2$ with $d(a, \tau) = \chi_2(a)(-1)^\tau$, $\lambda = 1$, by theorem 2).

Proof. Step 1 (d factors through (χ_2, τ)). Fix (a, τ) and compare the edges with valuations b and $b + 2$: they have the same parity increment and $S_{b+2}(a) = 4^{-1} S_b(a)$, so the displayed identity gives

$$d\big(4^{-1} S_b(a),\, \tau + b\big) = \lambda \chi(a) d(a, \tau) = d\big(S_b(a),\, \tau + b\big).$$

For fixed a, as b ranges over $\mathbb{Z}_{\geq 1}$ the residue $S_b(a)$ ranges over the coset $(3a+1)\langle 2 \rangle = U_{3^r}$ (since 2 is a primitive root mod 3^r), and for each fixed value $v = S_b(a)$ the residue class of b mod ϕ is determined, so letting τ vary gives both parities $\sigma = \tau + b$. Hence $d(4^{-1} v, \sigma) = d(v, \sigma)$ for *every* (v, σ): $d(\cdot, \sigma)$ is constant on cosets of $\langle 4 \rangle$. Since U_{3^r} is cyclic, $\langle 4 \rangle = \langle 2^2 \rangle$ is its unique index-2 subgroup, the squares $= \ker \chi_2$ (lemma 2). Therefore $d(a, \tau) = D(\chi_2(a), \tau)$ for some $D : \{\pm 1\} \times \mathbb{Z}/2 \to S^1$.

Step 2 (χ is trivial on the squares). Substituting into the identity and using $\chi_2(S_b(a)) = (-1)^b$ (theorem 1),

$$D\big((-1)^b,\, \tau + b\big) = \lambda\, \chi(a)\, D(\chi_2(a), \tau) \qquad \text{for all } a, b, \tau.$$

Fix b and τ: the left side does not depend on a, so $a \mapsto \chi(a) D(\chi_2(a), \tau)$ is constant on U_{3^r}. In particular χ is constant on the subgroup $\ker \chi_2$ of squares; a character constant on a subgroup equals 1 there. Hence χ factors through $U_{3^r} / \ker \chi_2 \cong \mathbb{Z}/2$, i.e. $\chi \in \{\mathbf{1}, \chi_2\}$. ☐

Lemma 3 (Irreducibility)**.** *The principal synchronized operator $\mathcal{L}_{\mathbf{1}}$ on $U_{3^r} \times (\mathbb{Z}/2)$ is irreducible (indeed primitive): every state is reachable from every state in exactly two steps.*

Proof. From any a, one step reaches any prescribed residue v (choose b with $2^{-b} = v/(3a + 1)$, possible since 2 generates U_{3^r}), and the parity increment of that step is determined by v alone:

$b \equiv [\chi_2(v) = -1] \pmod 2$, because $\chi_2(v) = (-1)^b$. Given a target (u, σ), choose the intermediate residue v with $\chi_2(v)$ set so that the two forced parity increments sum to $\sigma - \tau$; both square classes are nonempty, so this is always possible. □

Corollary 2 (Strict spectral gap at every finite level)**.** *For every $r \geq 2$, every tilt s, and every character $\chi \notin \{\mathbf{1}, \chi_2\}$ of U_{3^r},*

$$\rho\big(\mathcal{L}^{\mathrm{par}}_{s,Q,\chi}\big) < \rho\big(\mathcal{L}^{\mathrm{par}}_{s,Q,\mathbf{1}}\big),$$

where $\mathcal{L}^{\mathrm{par}}$ denotes the synchronized residue–parity operator. In particular $\kappa^{\mathrm{par}}_Q(s) > 0$ for every fixed $Q = 3^r$ and s.

Proof. $|\mathcal{L}_\chi| = \mathcal{L}_{\mathbf{1}}$ entrywise (the twist has unit modulus), and $\mathcal{L}_{\mathbf{1}}$ is irreducible (lemma 3). By Wielandt's equality theorem [39] (see also [14, Theorem 8.4.5]), $\rho(\mathcal{L}_\chi) = \rho(\mathcal{L}_{\mathbf{1}})$ would force $\mathcal{L}_\chi = e^{i\varphi} D\, \mathcal{L}_{\mathbf{1}} D^{-1}$ with D diagonal unimodular—which is precisely the identity hypothesised in theorem 3 with $\lambda = e^{-i\varphi}$. By theorem 3 this requires $\chi \in \{\mathbf{1}, \chi_2\}$, a contradiction. □

Remark. The proof of theorem 3 is purely structural: it uses only which edges exist (the richness of the branch set), not the values of the weights. It therefore holds simultaneously for all tilts and for any summable positive weighting of the valuations, and it bypasses the principal-unit functional equation entirely. This resolves the coboundary-classification question for the synchronized skew product; see section 9 for the height-strip refinement.

8 Conductor autonomy and collapse

Proposition 2 (Conductor reduction is autonomous)**.** *A character of conductor 3^c depends only on $a \bmod 3^c$. Since $3a \equiv 3(a \bmod 3^{c-1}) \pmod{3^c}$, the quantity $(3a+1)2^{-b} \bmod 3^c$ is a function of $(a \bmod 3^c, b)$ alone; hence the mod-3^c reduction of the dynamics is autonomous, and the gap ratio of a conductor-3^c sector is* independent of the ambient r *once $r \geq c$.*

Proof. Immediate from $3a \equiv 3(a \bmod 3^{c-1}) \pmod{3^c}$: the induced twisted operator on $(\mathbb{Z}/3^c)^\times \times$ (height) makes no reference to r. □

Autonomy gives an embedding of the level-3^c spectrum into the level-3^r operator. The following sharper statement says nothing else survives: the *entire* nonzero spectrum of a conductor-3^c sector is computed at level 3^c.

Theorem 4 (Conductor collapse)**.** *Let χ have conductor 3^c, $2 \leq c \leq r$. Then for the synchronized operator at level $Q = 3^r$ (residue–parity, or the height-strip operator with or without killing),*

$$\operatorname{spec}\big(\mathcal{L}^{(r)}_\chi\big) \setminus \{0\} = \operatorname{spec}\big(\mathcal{L}^{(c)}_\chi\big) \setminus \{0\}.$$

Consequently every gap ratio is exactly conductor-determined, and

$$\kappa_{3^r}(s) = 1 - \max_{2 \leq c \leq r} \gamma_c(s), \qquad \gamma_c(s) = \max_{\operatorname{cond}\chi = 3^c} \frac{\rho\big(\mathcal{L}^{(c)}_{s,\chi}\big)}{\rho\big(\mathcal{L}^{(c)}_{s,\mathbf{1}}\big)},$$

a maximum of finitely many r-independent constants. In particular κ_{3^r} is non-increasing in r, and (with corollary 2) $\kappa^{\mathrm{par}}_{3^r}(s) > 0$ for every fixed r; uniform positivity as $r \to \infty$ is equivalent to $\sup_c \gamma_c(s) < 1$.

Proof. Work with the Koopman form $M_\chi g(a,\cdot) = \chi(a)\sum_b w_b\, g(S_b(a),\cdot)$, where $(\cdot)$ stands for the auxiliary coordinate (parity τ, or the strip coordinate z with or without killing); the argument below acts only on the a-dependence and carries the auxiliary coordinate along unchanged. Here $A(a) = 3a+1$ and $S_b(a) = A(a)2^{-b}$. Every function of $(a,\cdot)$ expands uniquely as $g = \sum_{\eta\in\widehat{U_{3^r}}} \eta\otimes g_\eta(\cdot)$; the conductor of η is the least 3^j such that η factors through U_{3^j}, and we set

$$V_j = \Big\{\textstyle\sum_\eta \eta\otimes g_\eta :\ g_\eta = 0 \text{ unless } \operatorname{cond}\eta \le 3^j\Big\}, \qquad V_0\subset V_1\subset\cdots\subset V_r.$$

For a single character η,

$$M_\chi(\eta\otimes u) = \chi\cdot(\eta\circ A)\otimes(\text{transported } u), \qquad A(a) = 3a+1.$$

The function $\eta\circ A$ depends only on $a \bmod 3^{e-1}$ where $3^e = \operatorname{cond}\eta$ (since $3a \bmod 3^e$ depends only on $a \bmod 3^{e-1}$), so its Fourier support lies in conductors $\le 3^{e-1}$; multiplying by χ moves support into conductors $\le 3^{\max(c,e-1)}$. Hence

$$M_\chi V_j \subseteq V_{\max(c,\,j-1)}.$$

Thus V_c is invariant, and on the quotient filtration $V_c\subset V_{c+1}\subset\cdots\subset V_r$ the induced maps are strictly lowering, so M_χ is nilpotent on V_r/V_c (of index $\le r-c$). Therefore $\operatorname{spec}(M_\chi) = \operatorname{spec}(M_\chi|_{V_c})\cup\{0\}$, and $M_\chi|_{V_c}$ is the level-3^c operator by proposition 2. The principal radii agree at all levels ($M_{\mathbf 1}\mathbf 1 = \big(\sum_b w_b\big)\mathbf 1$), giving the ratio statement. □

Remark. theorem 4 is confirmed exactly by computation: on the residue–parity operator each γ_c is identical across all ambient levels tested ($Q\le 3^6$), e.g. $\gamma_2(0) = 0.654654$ and $\gamma_3(0) = 0.763817$ at every Q. The same collapse holds for the killed height-strip operator (the proof never touches the height index), which is why the strip conductor table of part III is r-independent.

9 Height-strip rigidity of the quadratic sector

On the discretised height strip (integer scale $R\ge 1$, increment $z' = z + c^* - Rb$ with $c^* = \lfloor R\log_2 3\rceil$, truncation to $z\in[0,L]$ with killing) the quadratic sector is not merely peripheral—its spectrum is a rigid rotation of the principal one.

Proposition 3 (Rotation identity)**.** *Let $D = \operatorname{diag}\big(\chi_2(a)\,e^{i\pi z/R}\big)$. Then on the height-strip operator, truncated or not, and for every tilt s,*

$$\mathcal L_{\chi_2} = e^{-i\pi c^*/R}\, D\,\mathcal L_{\mathbf 1}\, D^{-1}, \qquad \textit{hence} \qquad \operatorname{spec}\big(\mathcal L_{s,Q,L,\chi_2}\big) = e^{-i\pi c^*/R}\operatorname{spec}\big(\mathcal L_{s,Q,L,\mathbf 1}\big).$$

Proof. On an edge $(a,z)\to(a', z+c^*-Rb)$, using $\chi_2(a') = (-1)^b$ (theorem 1) and $e^{i\pi(c^*-Rb)/R} = e^{i\pi c^*/R}(-1)^b$,

$$\frac{d(a',z')}{d(a,z)} = \frac{\chi_2(a')e^{i\pi z'/R}}{\chi_2(a)e^{i\pi z/R}} = \chi_2(a')\,(-1)^b\,e^{i\pi c^*/R}\,\chi_2(a) = e^{i\pi c^*/R}\,\chi_2(a),$$

since $\chi_2(a')(-1)^b = 1$ and $\chi_2(a)^{-1} = \chi_2(a)$. This is the twisted coboundary identity with $\lambda = e^{i\pi c^*/R}$; as D is diagonal in (a,z) it commutes with the truncation to $z\in[0,L]$. □

χ_2 is part of the baseline dynamics, not an obstruction to equidistribution: its sector is a rigid rotation of the principal one, its Weyl sum is slaved to the valuation statistics, and it is quotiented into the reference law.

This explains the machine-precision equality $\rho(\mathcal{L}_{\chi_2}) = \rho(\mathcal{L}_{\mathbf{1}})$ observed on the killed strip for every (R, L) tested. For characters *outside* $\langle\chi_2\rangle$ no product-form strip potential exists: a potential $d(a, z) = \chi'(a)e^{i\omega z}$ satisfies the strip coboundary identity iff $\chi'(2)e^{i\omega R} = 1$ and $\chi'(3a+1) = \lambda'\chi(a)\chi'(a)$ on U_{3^r}; an exhaustive search over all character pairs for $r \le 4$ finds only $(\chi', \chi) = (\mathbf{1}, \mathbf{1})$ and (χ_2, χ_2). (The general—non-product-form—strip classification reduces to this principal-unit functional equation and is left open; corollary 2 settles the residue–parity case unconditionally.)

Part III

Numerical evidence (summary)

The full suite `collatz_tests` (ISO C11; FAST128/GMP backends; Arnoldi spectral solver) and Experiments A–I are documented in the companion [16]; every figure there is reproducible. We summarise only what bears on the statements above.

1. **Congruence (lemma 1).** Verified in 23,905,968/23,905,968 checks (seeds 3–10^6, $r \le 8$, native) and 144/144 in exact GMP arithmetic for a 16-digit seed at $r = 12$ with n up to 155 (values $\gg 2^{128}$).

2. **Key identity (theorem 1) and corollary 1.** $\chi_2(S(x)) = (-1)^b$ holds in $10^5/10^5$ tested seeds; $W_n(x; \chi_2, 0)$ has empirical mean $-0.353 \approx -\frac{1}{3}$ over 3.2×10^4 orbits, confirming the residues are biased $\frac{1}{3} : \frac{2}{3}$.

3. **Annealed laws.** Height tails fit $\log_2 \Pr(h \ge H) \approx -H$ (slope ≈ -0.97); high-ascent excursion bulk leans to the Cramér tilt $p^{(1)}$ $\big(D(\cdot \,\|\, p^{(1)}) = 0.093 \ll D(\cdot \,\|\, p) = 0.191$, boundary-excluded).

4. **Operator gaps.** On the residue-only operator every nontrivial sector is gapped; proposition 1 is reproduced to the digit. theorem 3 and corollary 2 are confirmed: on the residue–parity operator, a full-group scan finds exactly one peripheral sector ($k = \phi/2 = \chi_2$) at every level and tilt. theorem 4 is confirmed *exactly*: each γ_c is identical across all ambient levels. The computed sequences:

γ_c (residue–parity)	c=2	c=3	c=4	c=5	c=6
$s = 0$	0.6547	**0.7638**	0.7629	0.7202	0.7147
$s = 1$	0.9078	0.8691	**0.9264**	0.9210	0.9021

γ_c (killed strip, s=0)	c=2	c=3	c=4	c=5
	0.9069	0.8530	0.8261	0.7306

Both sequences are bounded away from 1 and peak at a *fixed small* conductor (boldface), decaying beyond it. Note the residue–parity sequence is *not* monotone in c (e.g. $\gamma_4 > \gamma_2$ at s=1), while the killed strip is monotone over the tested range: the extremal conductor depends on the operator, but bounded-conductor extremality is seen in both. See part IV.

5. **Shadow $\Rightarrow$ gap, only by tilt.** In a 1.25×10^6-window search, single-orbit "shadows" concentrate at the weakest-gap tilt $s=1$ (82% of flagged windows; 100% of long ones) and in weaker-gap sectors, but the per-sector signal is dominated by the smallest modulus—the circularity leak made explicit.

Remark (Provisional labels). The conclusions involving the labels `critical_C_A`, `high_ascent`, and `post_exit` (Experiments B, D, I) are *exploratory* until the formal C_A-critical certificate is fixed: the literal criterion $V^2 \le 2^A P$ is degenerate and the suite uses a working `height` definition. These results should not be read as theorem evidence in their current form.

Part IV

The refined H23a target

The finite-operator side is now largely a theorem. theorem 3 classifies the coboundaries; corollary 2 gives a strict gap at every finite level; theorem 4 reduces uniformity in Q to a single sequence of computable constants. What remains open is one statement about that sequence.

> This work identifies the correct operator (the faithful `sync` synchronized operator), *proves* the correct quotient ($\mathrm{Cob} = \langle\chi_2\rangle$, theorem 3), proves a strict non-coboundary gap at every finite level (corollary 2), and proves that the uniform gap is governed by the r-independent conductor sequence $\gamma_c(s)$ (theorem 4):
>
> $$\kappa_{3^r}(s) = 1 - \max_{2\le c\le r} \gamma_c(s), \qquad \gamma_c(s) < 1 \text{ for every } c.$$
>
> The single remaining finite-operator statement is that $\sup_c \gamma_c(s) < 1$.

Proof status.

- *Proved:* χ_2 is an exact coboundary (theorem 2); $\mathrm{Cob} = \langle\chi_2\rangle$ on the residue–parity skew product (theorem 3); strict gap $\rho(\mathcal{L}_\chi) < \rho(\mathcal{L}_{\mathbf{1}})$ at every finite level for every non-coboundary χ and tilt (corollary 2); conductor autonomy and collapse (proposition 2 and theorem 4); the height-strip rotation identity (proposition 3); the residue-only χ_2 ratio $\frac{2^{1+s}-1}{2^{1+s}+1}$ (proposition 1); one-step mode damping with the coboundary modes as the only undamped ones (lemma 4); the stability extraction step—near-peripheral spectrum yields a near-coboundary (proposition 4); *exact descent*—the primitive sector at level 3^c is isospectral to a multiplier-twisted operator one level down, so deep-digit comparisons are lossless (theorem 5 and corollary 4).
- *Proved (canonical witness):* the explicit cycle pair of theorem 17 has $W \equiv 4 \bmod 9$, giving full principal-unit generation and the uniform $\sqrt{3}$ cube-root separation for every primitive character—on the two-descent range unconditionally, and for all c granted carry depth.
- *Proved (pair-correlation engine):* the phase formula (proposition 7); the *exact nonstationary vanishing* lemma (lemma 5); the collision classification and mass identity (lemma 6); the endpoint obstruction—fixed-n operator-norm bounds are structurally false, so the assembly must run through spectral/trace quantities (proposition 8); and the trace identity $\mathrm{tr}((T^m)^*T^m) = \dim\Delta_s^m + \Theta_{m,c}$ with the thin remainder verified $\le 6\times10^{-4}$ and decreasing in c (lemma 7).

- *Proved (assembly bookkeeping):* the thin-sector bound (lemma 8), and—two further structural negative results—fixed-m chain powers bound the norm rather than the spectral radius (non-normal inflation), while the single trace saturates *exactly* at $\rho^2 \le 1$ at $s = 0$, with zero margin (proposition 9).

- *Proved (wrap-regime collision flattening):* the walk–window factorization $C_w = 2^{B_{m-1}} X_w$ (lemma 9) and the pigeonhole-flatness of the wrap-regime law, $\mathrm{Col}_m(c) \le \mathrm{poly}(c)\, 3^{-c}$ at $m \ge (1+\varepsilon)c$ (lemma 10; measured $3^c\,\mathrm{Col} \approx 2$): the deep digits of the window variable determine the suffix exactly (no wrapping of partial sums beyond scale $O(\log c)$). This is the renewal-flattening half of the wrap conjecture, in collision form.

- *Open (one lemma):* lift-phase chain cancellation (Conjecture 3): pure counting is once more exactly marginal, so the chain decay must come from the character phases carried by the source lifts $a_{j+1} \in a_j + 3^{c-m}\mathbb{Z}$ at each link. It implies $\limsup_c \gamma_c(s) \le \sqrt{\Delta_s} < 1$ in the limit (theorem 15), the high-conductor damping theorem with conjecturally sharp constant $\gamma_\infty(s) = \sqrt{(2^{1+s}-1)/(2^{1+s}+1)}$—matched by the observed tail at $s = 0$.

- *Proved (three-level descent):* two scalar descents plus one rank-2 descent (theorems 5 and 6, the latter verified to twelve digits)—a fixed dimension reduction by 27/2. *Disproved (rank saturation):* deeper carries depend on $b \bmod 2 \cdot 3^j$ and the internal classes refine by $\times 3$ per level; the nonzero-eigenvalue count grows $\approx \times 3$ per level (proposition 6), so descent stalls at depth three and there is *no* bounded-level reduction. The fixed-c_0 holonomy program is therefore limited to $c \le c_0 + 3$; the canonical witness and generation theorems hold on that range.

- *Open (short-witness coverage = the quantitative heart):* generation does not imply coverage; the uniform gap needs $\delta_c(L_0) \ge \delta_0$ at fixed witness length (Conjecture 5, strong $\Rightarrow$ constant gap; weak/polylog $\Rightarrow \kappa_Q \gg (\log Q)^{-2A}$, theorem 16). The controlled-length audit finds near-maximal coverage already at $L = 1$ on the audited range; the fixed-c_0 asymptotics await Task A. Comparison-graph expansion (Conjecture 6) remains an independent route to the polylog form (theorem 18).

- *Open (strip classification):* the non-product-form coboundary classification for the killed strip (the residue–parity case is settled; product-form strip potentials are excluded numerically for $r \le 4$).

- *Conditional:* a constant sync-quotient gap (corollary 3); a polylog gap under comparison-graph expansion (theorem 18).

Conjecture 1 (Bounded-conductor extremality)**.** *For each tilt $s \in [0,1]$ the sequence $\gamma_c(s)$ of theorem 4 attains its supremum at a fixed finite conductor: $\sup_{c\ge 2} \gamma_c(s) = \max_{c \le c_0} \gamma_c(s) < 1$. Numerically $c_0 = 3$ (s=0) and $c_0 = 4$ (s=1) for the residue–parity operator (through $c = 6$), and $c_0 = 2$ for the killed strip at $s = 0$ (through $c = 5$, where the sequence is monotone decreasing). Caution: monotonicity in c* fails *for the residue–parity operator ($\gamma_4 > \gamma_2$ at s=1), so the correct target is boundedness/extremality, not monotonicity.*

Corollary 3 (Conditional constant gap)**.** *Assume Conjecture 1. Then for all $Q = 3^r$ ($r \ge 2$) and $s \in [0,1]$,*

$$\rho(\mathcal{L}_{s,Q,\chi}) \le (1-\kappa_0(s))\, \rho(\mathcal{L}_{s,Q,\mathbf{1}}), \qquad \chi \notin \langle \chi_2 \rangle, \qquad \kappa_0(s) = 1 - \max_{c \le c_0} \gamma_c(s) > 0,$$

a constant *determined by finitely many computable operators. For the residue–parity operator the computed values give $\kappa_0(0) = 1 - 0.7638 = 0.2362$ and $\kappa_0(1) = 1 - 0.9264 = 0.0736$; for the killed strip* *at $s = 0$, $\kappa_0 = 1 - 0.9069 = 0.0931$. This is stronger than the fall-back target $\kappa_Q \gg 1/\mathrm{polylog}(Q)$, for which it suffices that $\gamma_c \le 1 - c^{-A}$ for some A.*

Proof. Immediate from theorem 4: $\kappa_{3^r} = 1 - \max_{2 \le c \le r} \gamma_c$, and under Conjecture 1 the maximum is attained at $c \le c_0$, independent of r. Each $\gamma_c < 1$ by corollary 2 applied at level 3^c. $\square$

10 Toward $\sup_c \gamma_c < 1$: stability of the classification

The route we consider most promising is a *quantitative* version of theorem 3: near-peripheral spectrum forces a near-coboundary, and the exact rigidity of the classification proof should then produce a contradiction for non-coboundary characters. Two ingredients of this program are proved here; the third is formulated precisely.

One-step mode damping

Write the Koopman form of the synchronized operator at level $Q = 3^c$ as $M_\chi g(a, \tau) = \chi(a) \sum_{b \ge 1} w_b\, g(S_b(a), \tau + b)$ with $w_b = 3^s q^b$, $q = 2^{-(1+s)}$, $W = \sum_b w_b = 3^s q/(1-q)$. Fourier modes are $\eta \otimes \epsilon$ with $\eta \in \widehat{U_Q}$ and $\epsilon \in \{0, 1\}$ the parity frequency. (Scope note: the classification theorem 3 is *weight-free*; the explicit multiplier below is special to the geometric branch weights $w_b \propto q^b$.)

Lemma 4 (One-step mode damping)**.** *For every mode,*

$$M_\chi(\eta \otimes \epsilon) = c_{\eta,\epsilon} \left[\chi \cdot (\eta \circ A)\right] \otimes \epsilon, \qquad c_{\eta,\epsilon} = \sum_{b \ge 1} w_b\, \zeta^b = \frac{3^s q\, \zeta}{1 - q\zeta}, \qquad \zeta = \eta(2)^{-1}(-1)^\epsilon,$$

so that

$$\frac{|c_{\eta,\epsilon}|}{W} = \frac{1-q}{|1 - q\zeta|} \;\le\; 1,$$

with equality iff $\zeta = 1$, i.e. iff $(\eta, \epsilon) \in \{(\mathbf{1}, 0), (\chi_2, 1)\}$—precisely the two coboundary modes (the constants and the potential $d(a, \tau) = \chi_2(a)(-1)^\tau$ of theorem 2). Quantitatively, if $\theta = \arg \zeta$ then $1 - |c|/W \asymp q\theta^2/(1-q)^2$ for small θ; the "swapped" modes $(\mathbf{1}, 1)$ and $(\chi_2, 0)$ have $\zeta = -1$ and damping ratio exactly $\frac{1-q}{1+q} = \frac{2^{1+s}-1}{2^{1+s}+1}$, the constant of proposition 1.

Proof. $\eta(S_b(a))(-1)^{\epsilon(\tau+b)} = \eta(3a+1)\,\eta(2)^{-b}(-1)^{\epsilon b}\,(-1)^{\epsilon\tau}$, and the b-sum is geometric. $\eta(2) = 1$ iff $\eta = \mathbf{1}$ and $\eta(2) = -1$ iff $\eta = \chi_2$, since 2 generates U_Q. $\square$

lemma 4 quantifies the mechanism: every non-coboundary mode is strictly damped in one step, at a rate governed by the angle of $\eta(2)$. It also exposes the difficulty: the least-damped primitive mode at conductor 3^c has $\theta = \pi/3^{c-1}$, hence one-step damping $\asymp 9^{-c}$; a proof of a *uniform* gap cannot ride a single mode and must use the spreading of Fourier mass under $\eta \mapsto \eta \circ A$ (whose coefficients are Jacobi-type sums with square-root cancellation [5]). This is the analytic heart that remains.

Near-peripheral spectrum forces a near-coboundary

Proposition 4 (Stability, extraction step)**.** *Let μ be an eigenvalue of M_χ with $|\mu| = (1-\varepsilon)W$, $\varepsilon \ge 0$, and let φ be an associated eigenfunction, $\psi = |\varphi|$, $u = \varphi/\psi$ on $\{\psi > 0\}$, $\lambda = \mu/|\mu|$. Let ν be* *the positive left Perron vector of $M_{\mathbf{1}}$. Define the nonnegative per-edge defect*

$$\delta(x,b) = w_b\Big[\psi(S_b x) - \mathrm{Re}\big(\overline{\lambda\chi(a)u(x)}\,\varphi(S_b x)\big)\Big], \qquad x=(a,\tau).$$

Then $\sum_b \delta(x,b) = \big(M_{\mathbf{1}}\psi - (1-\varepsilon)W\psi\big)(x) \ge 0$ *pointwise on* $\{\psi>0\}$*, and*

$$\sum_x \nu(x)\sum_b \delta(x,b) = \varepsilon W\,\langle \nu,\psi\rangle.$$

Consequently, for every $t>0$, outside an edge set of ν-weighted defect mass $\le \varepsilon W\langle\nu,\psi\rangle/t$ the unimodular potential $d=u$ satisfies the coboundary identity $d(S_b x)\approx \lambda\chi(a)d(x)$ with $w_b\,\psi(S_b x)\big(1-\cos(\text{phase error})\big)\le t$. In particular, comparing the branches b and $b{+}2$ of a common source yields approximate $\langle 4\rangle$-invariance, $d(4^{-1}v,\sigma)\approx d(v,\sigma)$, on high-mass edges—the quantitative form of Step 1 of theorem 3.

Proof. The eigen-equation gives $|\mu|\psi(x) = \mathrm{Re}\big(\overline{\lambda\chi(a)u(x)}\,\chi(a)\sum_b w_b\varphi(S_b x)\big) = \sum_b w_b\,\mathrm{Re}\big(\overline{\lambda\chi(a)u(x)}\varphi(S_b x)\big)\cdot|\chi(a)|$, whence $\sum_b\delta(x,b) = M_{\mathbf{1}}\psi(x) - |\mu|\psi(x)$, which is ≥ 0 pointwise and has the stated ν-mass since $\langle \nu, M_{\mathbf{1}}\psi\rangle = W\langle\nu,\psi\rangle$. The tail bound is Chebyshev's inequality; each defect term dominates $w_b\psi(S_b x)(1-\cos)$ of the phase mismatch. □

Exact descent: lossless propagation across deep digits

The deepest digits turn out to need no approximate propagation at all. The mechanism is that the level-3^c dynamics reads only $a \bmod 3^{c-1}$ (the same compression behind proposition 2), so states in a common fiber of $U_{3^c}\to U_{3^{c-1}}$ have *identical* branch sums.

Theorem 5 (Exact descent)**.** *Let χ be primitive of conductor 3^c, $c\ge 2$, and let M_χ be the synchronized operator at level 3^c with branch weights $w_b>0$. Define, on functions of $(\bar a,\tau)\in U_{3^{c-1}}\times\mathbb{Z}/2$,*

$$N_1 g(\bar a,\tau) = \chi\big(A(\bar a)\big)\sum_{b\ge 1} w_b\,\chi(2)^{-b}\,g\big(S_b(\bar a) \bmod 3^{c-1},\,\tau+b\big),$$

where $A(\bar a)=3\bar a+1$ is a well-defined element of U_{3^c} (it depends only on $\bar a \bmod 3^{c-1}$). Then

$$\mathrm{spec}(M_\chi)\setminus\{0\} = \mathrm{spec}(N_1)\setminus\{0\},$$

with eigenspaces in bijection via $\varphi(a,\tau)=\chi(a)\,g(a \bmod 3^{c-1},\tau)$. In particular the primitive-sector ratio γ at level 3^c equals $\rho(N_1)/W$, computed one level down.

Proof. Suppose $M_\chi\varphi=\mu\varphi$, $\mu\ne 0$. The branch sum $\Sigma(\bar a,\tau)=\sum_b w_b\varphi(S_b(a),\tau+b)$ depends only on $(\bar a,\tau)$, because $S_b(a) \bmod 3^c = A(\bar a)2^{-b}$ is a function of $(\bar a,b)$. Hence $\varphi(a,\tau)=\mu^{-1}\chi(a)\Sigma(\bar a,\tau) =: \chi(a)g(\bar a,\tau)$ *exactly*, at every state. Substituting this form of φ at the successors and using $\chi(S_b(a)) = \chi(A(\bar a))\chi(2)^{-b}$,

$$\mu\, g(\bar a,\tau) = \sum_b w_b\,\chi(S_b(a))\,g\big(S_b(a) \bmod 3^{c-1},\tau+b\big) = N_1 g(\bar a,\tau),$$

and $g\ne 0$ since $\varphi\ne 0$. Conversely, given $N_1 g=\mu g$ with $\mu\ne 0$, set $\varphi(a,\tau)=\chi(a)g(\bar a,\tau)\ne 0$; the same computation gives $M_\chi\varphi=\mu\varphi$. The two maps are mutually inverse on eigenspaces. □

Remark. Verified numerically: at $(Q,k,s) = (81,1,0),(81,1,1),(243,5,0)$ the nonzero spectra agree (top eigenvalues match to the eigenvalue condition number, ρ to ten digits); the complementary spectrum of M_χ is nilpotent, seen numerically as pseudospectral rings of radius $\approx 10^{-8}$.

Corollary 4 (Descent and the comparison graph)**.** *Every nonzero-eigenvalue eigenfunction satisfies* $\varphi(a,\tau)\chi(a)^{-1} = \varphi(a',\tau)\chi(a')^{-1}$ *whenever* $a \equiv a' \pmod{3^{c-1}}$*—the deep-digit comparisons of the conditional argument hold* exactly*, with zero defect, and the same factorization can be iterated: composing descents expresses the primitive sector at level* 3^c *as an explicitly phased operator at any level* 3^{c_0} *(the multiplier of the* j*-th descended operator is a product of* χ*-values along depth-*j *branches). Consequently the expansion needed in Conjecture 6 below is required only at a* bounded *level: what remains is that the accumulated unimodular multiplier system at a fixed level* 3^{c_0} *stays uniformly (in c) away from the coboundary class—a finite-dimensional non-degeneracy statement, since at fixed dimension "*δ*-far from every coboundary" implies a gap* $\kappa(\delta) > 0$ *by Wielandt's theorem and compactness.*

The descended multiplier and its holonomy

By theorem 5 the bounded-level object carrying all remaining information is the *accumulated multiplier*: after descending from level 3^c to level 3^{c_0}, each edge e of the fixed level-c_0 graph carries a phase $\chi(F(e))$ for an explicit, character-independent element $F(e) \in U_{3^c}$.

Proposition 5 (Two-descent multiplier recursion)**.** *With canonical least-residue lifts, the first descent gives* $F_1(x,b) = (3x+1)2^{-b} \in U_{3^c}$ *(the level-c lift of the successor), and the second descent gives*

$$F_2(\hat{x},b) = F_1(\hat{x},b)\,\rho^{\delta_b(\hat{x})}, \qquad \rho = 1 + 3^{c-1},$$

where $\delta_b(\hat{x}) \in \{0,1,2\}$ *is the top digit of the successor* $S_b(\hat{x})$ *at the upper level relative to the canonical lift of its reduction. Both steps are exact: the fiber ratio* $F_j(x+\delta 3^\ell,b)F_j(x,b)^{-1} = \rho^\delta$ *is independent of* (x,b) *for* $j \le 2$ *(verified exactly in computation). At depth* $j \ge 3$ *the fiber ratio acquires a parity twist—the successor's top digit shifts by a quantity depending on* $b \bmod 2$ *through* $2^{-b} \equiv (-1)^b \pmod 3$*—so the multiplier must be treated as an edge multiplier on the residue–parity graph; the resolution is the rank-2 descent below.*

Theorem 6 (Rank-2 parity-twisted descent)**.** *At the third descent the fiber ratio of the accumulated multiplier has the exact form*

$$\frac{m(\hat{a}+\delta 3^{L-1},b)}{m(\hat{a},b)} = \chi\big(r_0(\hat{a},\delta)\big)\cdot\omega^{\,\delta(-1)^b}, \qquad r_0(\hat{a},\delta) = 1+\delta 3^L(3\hat{a}+1)^{-1} \in 1+3^L\mathbb{Z}, \quad \omega = \chi(1+3^{c-1}),$$

with L *the current level. Splitting the branch sum by parity classes* $b \equiv \epsilon \pmod 2$ *and setting* $A_\epsilon(\hat{a},\tau) = \sum_{b\equiv\epsilon} w_b\, m(\hat{a},b)\, g(S_b\hat{a},\tau+\epsilon)$*, every* $\mu \neq 0$ *eigenfunction satisfies the exact fiber expansion* $\mu\, g(\hat{a}+\delta 3^{L-1},\tau) = \chi(r_0)\big[\omega^\delta A_0(\hat{a},\tau) + \omega^{-\delta}A_1(\hat{a},\tau)\big]$*, and the pair* (A_0,A_1) *satisfies a closed eigenproblem one level down:*

$$\mu A_\epsilon(\hat{a},\tau) = \sum_{b\equiv\epsilon(2)} w_b\, m(\hat{a},b)\,\chi\big(r_0(\hat{v}_b,\delta_b)\big)\big[\omega^{\delta_b}A_0(\hat{v}_b,\tau+\epsilon) + \omega^{-\delta_b}A_1(\hat{v}_b,\tau+\epsilon)\big],$$

with $\hat{v}_b$ *the reduced successor and* δ_b *its top digit. The correspondence is a bijection on nonzero eigenspaces: the primitive sector descends exactly onto a* rank-2 *operator on* $U_{3^{L-1}} \times \mathbb{Z}/2 \times \{0,1\}$*.*

Proof. The fiber-ratio formula is the computation in the proof sketch of proposition 5 made exact: the F_1-part of the ratio is $(3(\hat{a}+\delta 3^{L-1})+1)/(3\hat{a}+1) = r_0$, independent of b, and the accumulated ρ^{δ_b}-part shifts by $\rho^{\delta(-1)^b}$ because the successor moves by $\delta 3^{L-1}\cdot 3\cdot 2^{-b} \equiv \delta(-1)^b 3^L$ at the upper level. The fiber expansion follows by splitting the eigen-equation at the fiber point into parity classes; substituting it at the successors closes the system on (A_0, A_1), and the two maps (restriction to (A_0, A_1); reconstruction of g from the fiber expansion) are mutually inverse on nonzero eigenspaces, exactly as in theorem 5. Verified numerically at $(c, k, s) = (5, 1, 0)$: the full 324-dimensional primitive operator and the 24-dimensional rank-2 descended operator have equal spectral radius to twelve digits and matching significant spectra. $\square$

Proposition 6 (Negative result: rank saturation fails; descent stalls at depth three)**.** *An earlier draft conjectured that the internal rank stays* 2 *under all further descents.* This is false. *At the second rank-2 step the fiber variation of the accumulated* ρ^{δ_b}*-factors shifts the successor at* two *digit positions, through* $\delta\cdot 2^{-b} \bmod 9$*: the carry depends on* $b \bmod 6$*, not just on the parity* $b \bmod 2$*, and each further descent refines the internal classes by another factor of* 3*. The bounded-level reduction therefore does not continue. This is confirmed unconditionally by the nonzero-eigenvalue count of the full primitive operator (*$s = 0$*):*

c	4	5	6	7
$\#\{\lvert\lambda\rvert > 10^{-2}\}$	8	20	56	160
$4\phi_{c-3}$ *(three-level reduction)*	24	24	72	216

The count grows by a factor ≈ 3 *per level and is bounded by* $4\phi_{c-3}$*, exactly as predicted by a* fixed three-level *reduction (two scalar descents + one rank-2 descent, a constant dimension factor* $27/2$*) followed by no further collapse. Consequently the fixed-*c_0 *multiplier tables of the preceding discussion exist only for* $c \le c_0 + 3$*, and the bounded-level holonomy program (Conjecture 5 in its fixed-*c_0*,* $c \to \infty$ *reading) cannot be fed by descent: the witness audit below is valid on its stated range and does not extend asymptotically by this route. What survives in full: exact descent and the rank-2 step (theorems 5 and 6); the carry-depth property* on that range*; the canonical witness and full generation on that range; the classification, collapse, and strict-gap theorems, which never used bounded-level descent. The quantitative uniform-gap question reverts to the conductor sequence itself (Conjecture 1), for which the descent data adds new favourable evidence: the primitive-sector radii* decrease *in* c *(*0.7629, 0.6454, 0.6154, 0.6025 *at* k *small,* $s = 0$*,* $c = 4, \ldots, 7$*)—high conductor is increasingly damped, consistent with bounded-conductor extremality.*

The high-conductor damping problem

With bounded-level descent excluded, the remaining quantitative theorem is a *tail estimate*, attacked as an oscillation problem rather than a compactness problem.

Conjecture 2 (High-conductor damping)**.** *There are* c_1 *and* $\delta > 0$ *such that for every* $c \ge c_1$*, every primitive* χ *of conductor* 3^c*, and every* $s \in [0, 1]$*,* $\rho(M_\chi) \le (1-\delta)W$*. With the strict gap at each finite level (corollary 2) and conductor collapse (theorem 4), this is equivalent to* $\sup_c \gamma_c(s) < 1$ *(Conjecture 1).*

The available mechanisms are: the exact coboundary classification (theorem 3); one-step mode damping with equality only at the coboundary modes (lemma 4); stability extraction (proposition 4); three-level descent and the eigencount control (theorems 5 and 6 and proposition 6); and the numerically decreasing tail of γ_c.

In the character basis the operator has matrix $K[t_1, t_0] = c_{t_0,\epsilon}\, J(k - t_1, t_0)$, where

$$J(u, v) = \frac{1}{\phi} \sum_{a \in U_{3^c}} e_u(a)\, e_v(3a + 1)$$

are p-adic Jacobi sums [5, 15]. Two structural facts, computed exactly for $c \leq 7$: *(i)* conductor lowering makes each column of J supported on a fixed fraction 2/9 of the frequencies, with exact resonance columns ($|J| = 1$) at low conductor; *(ii)* the entrywise bound is useless—$\rho(|K|)/W = 1.85, 3.22, 5.58, 9.66$ at $c = 4, \ldots, 7$, growing by exactly $\sqrt{3}$ per level, while the true ratio is ≈ 0.6: at $c = 7$ a factor ≈ 16 of *phase cancellation* separates the truth from any magnitude argument. The gap is therefore a genuine oscillation phenomenon. The natural formulations are:

1. *Mode spreading / finite-time damping:* show that along the mode evolution $\eta \mapsto \chi \cdot (\eta \circ A)$ the near-undamped condition $\eta(2)^{-1}(-1)^\epsilon \approx 1$ cannot persist for m consecutive steps (with m absolute, or $m \ll c^A$ for polylog), using the phases of J, not only the magnitudes.

2. *Van der Corput / pair correlation:* bound $\|M_\chi^n\|_2$ for $n \asymp c$ via the word-pair correlation sums $\sum_a \chi\big(\Phi_w(a)/\Phi_{w'}(a)\big)$; stationarity of the phase ratios forces the exact coboundary relations, which are classified, so only the $\langle \chi_2 \rangle$ sector can resist—a quantitative use of theorem 3. The arithmetic input is p-adic stationary phase for the rational functions $\Phi_w/\Phi_{w'}$, where bounded (not square-root) cancellation suffices.

The pair-correlation engine

The second route is now equipped with its exact arithmetic. Both pieces below are rigorous; the first is a restatement of lemma 1's affine cocycle.

Proposition 7 (Pair-correlation phase formula)**.** *For a branch word $w = (b_0, \ldots, b_{n-1})$ set $B_j(w) = \sum_{i<j} b_i$ and*

$$C_w = \sum_{j=0}^{n-1} 3^{\,n-1-j} 2^{B_j(w)}.$$

Then the n-step map along w is $S_w(a) = (3^n a + C_w)\, 2^{-B_n(w)}$, and for two words w, w',

$$R_{w,w'}(a) = \frac{S_w(a)}{S_{w'}(a)} = 2^{\,B_n(w') - B_n(w)}\, \frac{3^n a + C_w}{3^n a + C_{w'}}, \qquad \frac{R'_{w,w'}}{R_{w,w'}}(a) = \frac{3^n\,(C_{w'} - C_w)}{(3^n a + C_w)(3^n a + C_{w'})}.$$

Since $C_w \equiv 2^{B_{n-1}} \not\equiv 0 \pmod 3$, the denominators are units on U_{3^c}, so the logarithmic derivative has constant 3*-adic valuation $d = n + v_3(C_w - C_{w'})$ with unit leading part. Stationarity of the pair phase is exactly high* 3*-adic divisibility of $C_w - C_{w'}$; note C_w does not depend on the last symbol b_{n-1}.*

Lemma 5 (Exact nonstationary vanishing)**.** *Let χ be primitive of conductor 3^c and let w, w' satisfy $d = n + v_3(C_w - C_{w'}) \leq c - 2$. Then*

$$\sum_{a \in U_{3^c}} \chi\big(R_{w,w'}(a)\big) = 0.$$

Proof. The argument is a Postnikov-type expansion of the character along principal units [26] (cf. [15, §12.3]). Write $L = (\log R)' = R'/R$, so $v_3(L(a)) = d$ with unit part $\mu(a) = 3^{-d}L(a)$, and $v_3(L') \geq d + n > d$. Split $a = a_0 + 3^{c-d-1}t$ with $t \in \mathbb{Z}/3^{d+1}$ (every lift of a unit is a unit). Taylor expansion of $\log R$ gives

$$R(a)/R(a_0) = 1 + 3^{c-1}\, t\, \mu(a_0)\, (1 + O(3)) + E,$$

where every higher term E has valuation $\geq 2(c-d-1) + d = 2c - d - 2 \geq c$ because $d \leq c - 2$. Hence $\chi(R(a)) = \chi(R(a_0))\, e\big(\kappa\, \mu(a_0)t/3\big)$ for a unit κ (χ primitive is nontrivial on $1 + 3^{c-1}\mathbb{Z}$). For each fixed a_0 the inner sum over $t \in \mathbb{Z}/3^{d+1}$ runs over full periods of a nontrivial third root of unity ($\mu(a_0)$ is a unit), so it vanishes identically. $\square$

Verified exactly: for $n = 2, 3$ and $c = 6, 7$ the sampled pair sums are machine zeros precisely on the predicted range $n + v_3(C_w - C_{w'}) \leq c - 2$, with structured values ($\frac{1}{2}$, 1) only in the near-stationary transition zone.

Lemma 6 (Collision classification and collision mass)**.** *C_w depends only on the prefix $(b_0, \ldots, b_{n-2})$, via the recursion $C_w = 3C_{w^-} + 2^{B_{n-1}}$. Conversely, distinct prefixes give distinct constants: if $B_{n-1}(w) = B_{n-1}(w')$ this follows by induction ($3(C_{w^-} - C_{w'^-}) = 0$ forces prefix equality); the unequal-B_{n-1} case is excluded by 2-adic digit analysis of $2^B(2^\Delta - 1) = 3(C_{w^-} - C_{w'^-})$ (the left side has 2-adic valuation $B_{n-1} \geq n-1$, forcing implausibly deep 2-adic agreement of the prefixes), verified exhaustively for $n \leq 5$, $b_i \leq 12$: the number of distinct constants is exactly 12^{n-1}. Consequently the exactly-stationary pair mass is*

$$\sum_C \Big(\sum_{w:\, C_w = C} p_s(w) \Big)^2 = \Delta_s^{n-1}, \qquad \Delta_s = \sum_b p_s(b)^2 = \frac{2^{1+s} - 1}{2^{1+s} + 1},$$

*confirmed numerically to truncation accuracy (*0.05–0.2% *at $B = 12$).*

Assembly via trace powers

Two structural facts dictate the shape of the assembly.

Proposition 8 (Endpoint obstruction: fixed-n norm bounds fail)**.** *A matrix entry of $(T^n)^* T^n$ constrains the source a to a coset of $1 + 3^{c-n}\mathbb{Z}$, on which every nonstationary pair phase is* constant *(its logarithmic derivative has valuation $d \geq n$, and $(c - n) + d \geq c$). Hence lemma 5 produces no within-entry cancellation, and the fixed-n bound $\|T^n\|^2 \leq \Delta_s^{n-1} + o_c(1)$ is* false*: at $s = 0$, $\|T^2\|^2 \approx 1.63$ versus $\Delta_0 = 0.33$ (stable in c). The operator norm of a fixed power is inflated by the non-normal (nilpotent) structure; only spectral quantities can obey the Δ_s scaling.*

Lemma 7 (Trace identity, main term)**.** *For fixed m and B, and c large,*

$$\mathrm{tr}\big((T^m)^* T^m\big) = \dim \cdot \Delta_s^m \; + \; \Theta_{m,c},$$

where the main term comes from word pairs with coincident endpoints and equal 2-power parts: $S_w(a) = S_{w'}(a)$ with $B_m(w) = B_m(w')$ forces $C_w = C_{w'}$ with the source a ranging over all *of U_{3^c}; by lemma 6 such pairs agree in the first $m - 1$ symbols, hence (the path phase $\Psi_w(a) = \prod_{i<m} \chi(x_i)$ being blind to the last symbol) have* identical *path phases, contributing mass exactly Δ_s^m per state; equal B_m then forces $w = w'$. The thin remainder $\Theta_{m,c}$ collects pairs with $2^{B_m(w)} \neq 2^{B_m(w')}$, whose source a is confined to lower-dimensional fibers. Numerically $\Theta_{m,c}/(\dim \Delta_s^m) \leq 6 \times 10^{-4}$ for $m \leq 4$ at $c = 6$,* decreasing *in c ($\leq 2 \times 10^{-4}$ at $c = 7$), and the $m = 1$ identity is exact.*

Lemma 8 (Thin-sector bound). *In the no-wrap regime (branch cutoff B, word length m with $C_w < 3^c$ for all truncated words), the thin remainder of lemma 7 satisfies*

$$\Theta_{m,c} \;\le\; C\, m^2 B \left(\tfrac{3}{2}\right)^m \cdot 3^m, \qquad \textit{hence} \qquad \frac{\Theta_{m,c}}{\dim \Delta_0^m} \le C\, m^2 B \left(\frac{9}{2}\right)^m 3^{-c} \xrightarrow[c\to\infty]{} 0 \quad \textit{for } m \le 0.7\, c.$$

Proof. A thin pair has $2^{B(w')} \neq 2^{B(w)}$; the endpoint equation $3^m a\,(2^{B'} - 2^B) = C_{w'} 2^B - C_w 2^{B'}$ has unit solutions only if $v_3(\mathrm{RHS}) = m + v$, $v = v_3(2^\Delta - 1) \le 1 + \log_3(mB)$, and then a is confined to a class with at most 3^{m+v} unit solutions. Solvability constrains $C_{w'} \equiv C_w 2^{\,B'-B} \pmod{3^{m+v}}$; in the no-wrap regime the collision classification (lemma 6) puts at most one prefix in each class, of mass $\le (1-r)^{m-1} \le 2^{-(m-1)}$ at $s = 0$. Summing over the $O(mB/3^v)$ values of Δ at each valuation v and bounding all phases by 1 gives the display; the comparison with $\dim \Delta_0^m = 4 \cdot 3^{c-1-m}$ is immediate. ☐

Proposition 9 (Two structural obstructions to the naive assembly). (a) Fixed-m chain powers fail. *For fixed m, $\mathrm{tr}\big[((T^m)^* T^m)^k\big]^{1/k} \to \|T^m\|^2$ as $k \to \infty$, and the non-normal inflation makes $\|T^m\|^2 > 1$ for small m ($\|T^2\|^2 \approx 1.62$, $\|T^3\|^2 \approx 1.15$, stable in c): chain powers at fixed m bound the* norm, *not the spectral radius, and yield no gap (verified: $\mathrm{tr}(P_2^k) = 324$, 350, 402 at $c = 7$, flat in k).* (b) The single trace is exactly marginal. *From $\rho^{2m} \le \mathrm{tr}((T^m)^* T^m) = \dim \Delta_s^m (1 + o(1))$ one gets $\rho^2 \le 3^{\,c/m} \Delta_s (1 + o(1))$, optimal at the largest admissible m; the pigeonhole floor (collision mass $\ge\asymp 3^{-c}$) blocks $m \gg c$, and at $m \asymp c$ the bound saturates at $\rho^2 \le 3\Delta_s$—which at $s = 0$ equals* 1 exactly. *The structural coincidence $\Delta_0 = 1/3$, $\dim \asymp 3^c$ leaves zero margin: no single-trace argument can produce the gap.*

The two obstructions intersect to locate the true remaining problem: the trace analysis must run in the *wrap regime*, with $m \asymp c$ *and* chain order $k \ge 2$, where the word constants wrap modulo 3^c and collisions are governed by the equidistribution of C_w mod 3^c rather than by integer injectivity.

The wrap regime: collision flattening (proved) and lift-phase cancellation (open)

The wrap-regime analysis splits into a distributional half, which we can prove, and a chain-phase half, which we localize precisely.

Lemma 9 (Walk–window factorization). *For any word w of length $m \ge c$,*

$$C_w \equiv 2^{\,B_{m-1}}\, X_w \pmod{3^c}, \qquad X_w = \sum_{k=0}^{c-1} 3^k\, 2^{-(b_{m-1-k} + \cdots + b_{m-2})} \quad (k \ge 1), \quad X_w \equiv 1 + 3\mathbb{Z},$$

i.e. the constant is a unit multiple of a window variable *depending only on the last $c-1$ symbols, by the same reversal as the sliding-window congruence (lemma 1). The walk factor $2^{B_{m-1}}$ and the window are coupled only through the last $c-1$ increments.*

Lemma 10 (Wrap-regime collision flattening). *At $s = 0$, for $m \ge (1+\varepsilon)c$,*

$$\mathrm{Col}_m(c) = \Pr\big[C_w \equiv C_{w'} \bmod 3^c\big] \;\le\; K(c)\, 3^{-c}, \qquad K(c) = O(\mathrm{poly}(c)),$$

and numerically $3^c\, \mathrm{Col}_m(c) \approx 2$ at $m = 2c$ ($1.75, 1.81, 1.88, 1.93, 1.98$ for $c = 4, \ldots, 8$), ≈ 4 at $m = c$: the wrap-regime law of C_w is pigeonhole-flat up to a small constant.

Proof sketch. By lemma 9 and Cauchy–Schwarz (conditioning on the walks and one window), $\mathrm{Col}_m(c) \le \max_\rho \Pr[X'/X = \rho] \le \mathrm{Col}(X)$. For the window variable, the digit at scale 3^k constrains the backward partial sum s_k modulo $2 \cdot 3^{k-1}$; since $s_k \le \max b \cdot c = O(c)$, there is *no wrapping* once $2 \cdot 3^{k-1} > s_{\max}$, i.e. for all $k \ge k_0 = O(\log_3 c)$: the deep digits determine the partial sums—hence the suffix symbols—*exactly.* Fibers of X are therefore single suffixes up to the $O(\log c)$ shallow digits, and

$$\mathrm{Col}(X) \le 3^{O(\log c)} \sum_{\text{suffixes}} p(\cdot)^2 = \mathrm{poly}(c)\, \Delta_0^{c-O(\log c)} = \mathrm{poly}(c)\, 3^{-c+O(\log c)}. \qquad \square$$

This proves the renewal-flattening half of the wrap conjecture in its L^2 (collision) form: the distribution of $C_w \bmod 3^c$ offers no collision excess for the chain trace to exploit against us. What it does *not* yet provide is the gap itself: a pure counting bound on $\mathrm{tr}(P_m^k)$—keeping only collided links and bounding all phases by 1—is once again *marginal* (the per-link factor $3^m \cdot$ per-target mass is $\asymp K \ge 1$ at every m, by the same zero-margin coincidence as proposition 9). The decay seen in the measured chain traces must therefore come from the *lift phases*: at each chain link the source lifts $a_{j+1} \in a_j + 3^{c-m}\mathbb{Z}$ carry character phases through the next link's words, and these lift sums cancel (vanishing-lemma style, in the lift variable) except on an exponentially thin set. The remaining statement is:

Conjecture 3 (Lift-phase chain cancellation; *superseded*)**.** *For $m \asymp c$ and some $k_0 \ge 2$, the chain trace with lift-phase cancellation included satisfies $\mathrm{tr}\big[((T^m)^*T^m)^{k_0}\big] \le 3^{\,c-\epsilon m k_0+o(c)}$ for some $\epsilon > 0$, uniformly over primitive χ. Together with lemma 10 this is the full wrap-regime input.* Status: superseded as the live finite-side input. The threshold cascade route (lemma 14 and proposition 13) requires neither chain traces nor lift-phase cancellation; its single analytic input is the threshold flattening statement theorem 10. This conjecture and theorem 15 below are retained as documentation of the earlier trace-side route.

Theorem 7 (Lift-sum factorization)**.** *In the chain-trace expansion of* $\mathrm{tr}[((T^m)^*T^m)^k]$*, fix the words and the base source a_1. The link constraints give $a_{j+1} = a_1 + D_j + 3^{c-m}T_j$ with D_j word-determined and $T_j = t_1 + \cdots + t_j$ the cumulative lifts; the chain phase collapses to a product over the* adjacent *word pairs at the sources,*

$$\prod_{j=1}^{k} \chi\big(R_j(a_j)\big), \qquad R_j = \textit{pair-correlation phase of } (w_{j-1}, w_j') \textit{ (the } i{=}0 \textit{ factors cancel),}$$

and since a_j depends only on T_{j-1} and the map $(t_1, \ldots, t_{k-1}) \mapsto (T_1, \ldots, T_{k-1})$ is triangular, the lift sum factorizes exactly *into independent one-variable sums:*

$$\sum_{\mathbf{t} \in (\mathbb{Z}/3^m)^{k-1}} \prod_j \chi\big(R_j(a_j)\big) = \chi(R_1(a_1)) \prod_{j=2}^{k} \sum_{T \in \mathbb{Z}/3^m} \chi\big(R_j(\alpha_j + 3^{c-m}T)\big).$$

The multivariate lift-phase problem is therefore univariate*: a coset character sum of a pair-correlation phase, per link.*

Lemma 11 (Coset dichotomy)**.** *Let R be a pair-correlation phase, χ primitive mod 3^c, and let $d = v_3\big((\log R)'(\alpha)\big)$ be the* pointwise *valuation at the coset base. Then (under the window $2(c-m)+d+1 \ge c$)*

$$\sum_{T \bmod 3^m} \chi\big(R(\alpha + 3^{c-m}T)\big) = \begin{cases} 0, & d \le m-1, \\ 3^m\, \chi(R(\alpha)), & d \ge m. \end{cases}$$

There is no intermediate boundary behaviour: at $d = m-1$ the Postnikov [26] linear coefficient $\lambda_1 = 3^{-(c-1)}\bigl[\log R(\alpha + 3^{c-m}) - \log R(\alpha)\bigr]$ is a unit by definition of d, and the full-period mod-3 sum vanishes. Verified exhaustively: at $(c, m, n) = (8, 4, 4)$ and $(7, 3, 3)$, all 1,734 sampled configurations with pointwise $d = m-1$ have $\lambda_1 \in \{1, 2\}$ and machine-zero sums; all non-vanishing cases have $\lambda_1 = 0$, i.e. true valuation $\geq m$. (An earlier draft reported a boundary anomaly; it was an artifact of classifying by the nominal *valuation $\min_i[i + v_3(\Delta C_i)]$—degenerate minima, where the leading units cancel mod 3, raise the true valuation in $\approx 16\%$ of nominal boundary cases. The degeneracy costs one mod-3 condition per level, so the migrated mass is geometrically controlled.)*

Proposition 10 (Trace-method ceiling)**.** *By theorem 7 and lemma 11, a chain survives only if every adjacent pair is true-stationary ($v_3((\log R_j)') \geq m$), which prefix-locks the words up to geometrically thin degenerate-minima corrections; the surviving phases are then* exactly constant *per coset, and two further exact resources apply—the base-point sum over a_1 (a complete U-sum, vanishing unless $v_3(\sum_j L_j) \geq c-1$) and the cyclic closure constraint (worth $\mathrm{poly} \cdot 3^{-(c-m)}$ by collision flattening). Carrying out the complete accounting with all of these, every variant lands at bounds of the form $\rho^2 \leq 3^{\,1/m} \cdot (const/\sqrt{m})^{1/m} \to 1$: strictly below 1 at every finite level—re-deriving the strict gap—but with no uniform δ. Since the coset weights are exactly $\{0, 1\}$ (no intermediate values; lemma 11), the boundary-transition-operator route degenerates, and the conclusion is structural:* the trace method has a marginality ceiling. *The surviving deep-stationary chains carry unimodular constant phases and trace mass $\Theta(3^k)$, which the inequality $\rho^{2mk} \leq \mathrm{tr}$ cannot convert into a uniform gap—the trace mass is carried by many small eigenvalues, not by ρ. A proof of $\sup_c \gamma_c < 1$ therefore requires a non-trace mechanism: either direct operator-norm control of $\|T^m g\|$ on the structured complement of the deep-stationary sector (an eigenvector-aware estimate), or the finite-Q certificate route of the prior program. The numerically observed limit $\gamma_\infty(s) = \sqrt{\Delta_s}$ stands as the conjectural sharp constant.*

The live route: eigenvector-aware norm control

Three measurements settle how the non-trace route must run. Let P_{stat} project onto the modes of conductor $\leq 3^{c-m}$ (the functions constant on the lift cosets—the deep-stationary directions).

(i) Top eigenvectors avoid the stationary sector. At $(c, k) = (6, 1), (7, 5)$ the top right eigenvector carries mass only 0.01–0.11 on $\mathcal{H}_{\mathrm{stat}}$ for $m = 2, 3, 4$: the trace ceiling is produced by spectrally irrelevant small modes, exactly as the ceiling analysis predicted.

(ii) The compressed stationary operator vanishes—exactly. The measurement $\rho\bigl(P_{\mathrm{stat}} T^m P_{\mathrm{stat}}\bigr)^{1/m} \leq 3 \times 10^{-4}$ that motivated the block decomposition is in fact machine roundoff of an exact identity:

Theorem 8 (Conductor raising; exact stationary ejection)**.** *Let χ be primitive modulo 3^c and let P_{top} project onto the modes $\eta \otimes \varepsilon$ with η of conductor exactly 3^c. Then*

$$P_{\mathrm{top}}^{\perp}\, T_{\chi,c,s} = 0 :$$

the image of the synchronized Koopman operator lies entirely in the top-conductor sector. Consequently, for every threshold $j < c$, writing $P_{\leq j}$ for the projection onto modes of conductor $\leq 3^j$,

$$P_{\leq j}\, T^m = 0 \qquad (m \geq 1),$$

so in particular $P_{\mathrm{stat}} T^m P_{\mathrm{stat}} = 0$ identically; and $\mathrm{spec}(T) \setminus \{0\} = \mathrm{spec}(P_{\mathrm{top}} T P_{\mathrm{top}}) \setminus \{0\}$: all spectral action takes place in the primitive sector, of dimension $8 \cdot 3^{c-2}$ (two thirds of the space).

Proof. It suffices to check the spanning modes $g = \eta \otimes \varepsilon$. By the synchronization rule,

$$(Tg)(a, \tau) = \chi(a)\,\eta(3a+1)\,\varepsilon(\tau) \cdot \sum_{b \geq 1} w_b\,\eta(2)^{-b}\varepsilon(b),$$

so up to a scalar the output is the function $a \mapsto \chi(a)\,\eta(3a+1)$ (times a parity character). Since $3a+1 \bmod 3^c$ depends only on $a \bmod 3^{c-1}$, the function $a \mapsto \eta(3a+1)$ factors through the reduction $U_{3^c} \to U_{3^{c-1}}$ and hence lies in the span of the multiplicative characters ζ of conductor $\leq 3^{c-1}$ (these are exactly the functions invariant under $\ker(U_{3^c} \to U_{3^{c-1}})$, by dimension count). For any such ζ, the restriction of $\chi\zeta$ to that kernel equals the restriction of χ, which is nontrivial because χ is primitive; hence $\mathrm{cond}(\chi\zeta) = 3^c$ exactly. Every Fourier component of the output therefore has top conductor. The corollaries follow from $P_{\leq j} P_{\mathrm{top}} = 0$ and $\mathrm{im}\, T \subseteq \mathcal{H}_{\mathrm{top}}$. ☐

Remark. The identity is one-sided: $TP_{\mathrm{top}}^{\perp} \neq 0$ (measured norm ≈ 1.16)—low-conductor modes feed the primitive sector and never return. In transfer (pushforward) form the statement transposes to $MP_{\leq j} = 0$; both forms were verified to 10^{-13} at $c = 5, 6$, together with the exact agreement of $\mathrm{spec}(T)$ and $\mathrm{spec}(P_{\mathrm{top}} T P_{\mathrm{top}})$ to all computed digits. This settles the block-decomposition corner exactly ($A = 0$, not merely small), and it explains measurement (i) structurally: the deep-stationary trace mass counted by the ceiling (proposition 10) lives in directions that T ejects in a single step and that no power ever revisits.

(iii) The norm decay rate is empirically c-uniform at $m \gtrsim c$.

$\sigma_1(T^m)^{1/m}$	m=4	m=8	m=12	m=16	m=20
$c = 5$	0.9647	0.8088	0.7470	0.7199	0.7044
$c = 6$	0.9626	0.8373	0.7655	0.7284	0.7087
$c = 7$	0.9644	0.8658	0.7782	0.7302	0.7025

At $m = 20$ the rate is flat in c to $\pm 0.6\%$. The structural reason is exact: for $m \geq c$ the endpoint obstruction of proposition 8 *evaporates.*

Proposition 11 (Complete-kernel formula)**.** *Let $m \geq c$. Then $3^m \equiv 0 \pmod{3^c}$, so the m-step endpoint $S_w(a) \equiv C_w 2^{-B_m(w)} \pmod{3^c}$ is determined by the word alone, independently of the source a. Consequently the kernel of T^m factorizes as*

$$K_m\big((x,\sigma),(a,\tau)\big) = \sum_{\substack{w:\, C_w 2^{-B_m(w)} \equiv x \ (3^c) \\ B_m(w) \equiv \sigma - \tau \ (2)}} p_s(w)\,\Psi_w(a), \qquad \Psi_w(a) = \prod_{i=0}^{m-1} \chi\big(x_i(a,w)\big),$$

and the entries of $H_m = (T^m)^* T^m$ are complete word-pair sums*:*

$$H_m\big((a,\tau),(a',\tau')\big) = \sum_{\substack{(w,w'):\, C_w 2^{-B_m(w)} \equiv C_{w'} 2^{-B_m(w')} \ (3^c) \\ B_m(w) \equiv B_m(w') + \tau' - \tau \ (2)}} p_s(w)\,p_s(w')\,\overline{\Psi_w(a)}\,\Psi_{w'}(a').$$

The common-endpoint constraint selects word pairs only—no source coset appears—so the incomplete-coset obstruction of proposition 8 is structurally absent in this regime, and the Schur test [31] becomes available.

Row-sum audit (Schur sharpness). For the unweighted Schur bound $\|T^m\|^2 \le R_m := \sup_\xi \sum_{\xi'} |H_m(\xi,\xi')|$ (states $\xi=(a,\tau)$), at $s=0$ and $m=c,\ c+2,\ 2c$:

	$c=5$			$c=6$			$c=7$		
m	5	7	10	6	8	12	7	9	14
$\sigma_1(T^m)^{1/m}$	0.9231	0.8364	0.7706	0.9096	0.8373	0.7655	0.9040	0.8363	0.7443
$R_m^{1/(2m)}$	0.9547	0.8600	0.7849	0.9350	0.8567	0.7823	0.9280	0.8620	0.7676

The Schur bound tracks $\sigma_1^{1/m}$ within 3% at every (c,m), and singular-vector weights move it by $\le 1\%$ in either direction: the *unweighted* row sum is essentially sharp, so the Schur norm is the correct vehicle and no weighted refinement is forced. At $m=2c$ the bound itself descends in c (0.7849, 0.7823, 0.7676). The effective constant $A_m(c)=R_m/\Delta_s^m$ of the anticipated shape $R_m \le A_m\Delta_s^m$ satisfies $A_m^{1/(2m)} = 1.61,\ 1.49,\ 1.33$ at $c=7$, $m=7,9,14$ (descending): the row-multiplicity growth rate is $\approx 1.8^m$, comfortably below the critical rate 3^m at which the bound $\|T^m\|^{1/m} \le A_m^{1/(2m)}\sqrt{\Delta_s}$ becomes vacuous at $s=0$.

Degenerate-minimum mass audit (the class-IV test). The proposed row-sum decomposition treats by absolute value the class of word pairs whose *nominal* nonstationarity $d_{\text{nom}} = \min_i[i+v_3(\Delta C_i)] \le c-2$ degenerates, through mod-3 cancellation among the leading terms, to true pointwise valuation $d_{\text{true}}(a) \ge c-1$—escaping the vanishing lemma. A Monte Carlo audit (2×10^6 pairs per configuration, $c=5,6,7$, $m=c,\ c+2,\ 2c$; classifier validated against exact complete sums: machine zeros for nondegenerate pairs, $|S|/\varphi \in [0.5,1]$ for all classified-degenerate pairs) gives a sharp two-sided verdict. The *level-cost mechanism is confirmed exactly*: $\Pr[d_{\text{true}} - d_{\text{nom}} \ge k]$ decays by a factor 3 per level at every (c,m). But the *aggregate mass lemma fails in the large-m regime*: the degenerate mass is flat in m (degeneration is a prefix-local phenomenon), and its conditional rate among endpoint-matched pairs rises from $\approx \frac{1}{7}$ of the unconditional rate at $m=c$ to full decorrelation at $m=2c$. Consequently the absolute-value bound on the class, $2\varphi\cdot\Pr[\text{matched}\wedge\text{degenerate}]$, is *not* $O(\rho^m)$: at $m=2c$ it exceeds the entire measured row sum by factors 14, 31, 77 for $c=5,6,7$. In the regime $m=c$, however, endpoint matching *suppresses* degeneracy (the window and the prefix overlap), and the class-IV mass is only 2–4% of the measured row sum, stable-to-decaying in c. The strategic consequence: the Schur bookkeeping must be run at $\alpha=1$ (i.e. $m=c$, the completeness threshold), where a pure mass treatment of the degenerate class empirically suffices and the row-sum rate to beat is $R_c^{1/2c} = 0.9547,\ 0.9350,\ 0.9280$ (descending in c); at $\alpha\ge 2$ the degenerate class would itself require phase cancellation.

The adjoint-side kernel and the threshold audit. The bookkeeping is run on the adjoint-side kernel $G_m = T^m(T^m)^*$, whose entries carry *complete source sums*: for states $\xi=(x,\sigma)$, $\xi'=(x',\sigma')$,

$$G_m(\xi,\xi') = \sum_{\substack{w\to x,\ w'\to x' \\ B_m(w)-B_m(w')\equiv\sigma-\sigma'\ (2)}} p_s(w)\,p_s(w')\,\chi(2)^{\sum_i(B_i'-B_i)} \sum_{a\in U_{3c}} \chi\Big(\prod_{i=1}^{m-1} \frac{3^i a + C_i(w)}{3^i a + C_i(w')}\Big),$$

so the coset dichotomy applies *pair by pair*: class I (nonstationary nondegenerate pairs) contributes exactly zero to every entry. In a full word enumeration at $c=4,5$ ($m=c$, sup row) all 5.6×10^5 resp. 8.5×10^6 class-I pairs give machine zeros, and the four-class absolute mass reproduces the matrix row sum from above with $\approx 20\%$ internal cancellation margin. The G-form row rates at $m=c$ are

$$R_G^{1/(2c)} = 1.0743,\ 1.0294,\ 1.0003,\ 0.9854,\ 0.9720 \qquad (c=4,\dots,8),$$

descending and below 1 from $c = 7$. The exact class decomposition identifies the controlling class: stationarity at threshold is the *congruence cascade* $v_3(\Delta C_i) \geq c - 1 - i$ for every interior i, equivalently the nested conditions $B_i \equiv B_i' \pmod{2 \cdot 3^{c-2-i}}$, one factor of 3 deeper per level of c. The twisted near-locked pairs satisfying this cascade (class II) carry 58% resp. 48% of the sup-row mass at $c = 4, 5$ with per-class rate $1.0373 \to 0.9762$ (below 1 from $c = 5$); the locked class III is second ($0.9883 \to 0.9618$); the degenerate class IV is subordinate (0.8406); the untwisted remainder is negligible (0.60).

The near-locked cascade bound

The class-II combinatorics announced above can be made exact and proved. Write $t_i = B_i(w) - B_i(w')$ and $D_i = 2^{B_i} - 2^{B_i'}$, so that $\Delta C_1 = 0$ and $\Delta C_{i+1} = 3\Delta C_i + D_i$.

Theorem 9 (Cascade equivalence). *For a pair of c-words, the threshold stationarity cascade $v_3(\Delta C_i) \geq c - 1 - i$ (all interior i) holds* if and only if

$$B_i \equiv B_i' \pmod{2 \cdot 3^{c-3-i}}, \qquad 1 \leq i \leq c - 3.$$

Proof. By lifting-the-exponent (a standard p-adic valuation identity; see, e.g., [13]), $v_3(2^t - 1) = 0$ for t odd and $= 1 + v_3(t)$ for t even; since the multiplicative order of 2 modulo 3^k is $2 \cdot 3^{k-1}$, $v_3(D_i) \geq k \iff t_i \equiv 0 \pmod{2 \cdot 3^{k-1}}$. Forward: if the cascade holds at i and $i + 1$, then $v_3(D_i) = v_3(\Delta C_{i+1} - 3\Delta C_i) \geq \min(c - 2 - i,\, c - i) = c - 2 - i$ for $i \leq c - 3$. Converse: $\Delta C_2 = D_1$ has $v_3 \geq c - 3$, and inductively $v_3(\Delta C_{i+1}) \geq \min(1 + v_3(\Delta C_i),\, v_3(D_i)) \geq c - 2 - i$. ☐

Lemma 12 (Cascade mass bound). *For independent p_0-random words,*

$$3^{-(c-3)} \;\leq\; \Pr[\text{cascade}] \;\leq\; K \cdot 3^{-(c-3)}, \qquad K = 1 + 4 \sum_{k \geq 0} 3^k 2^{-2 \cdot 3^k} < 2.2.$$

Moreover $\sup_w \Pr_{w'}[\text{cascade} \mid w] \leq (2/3)^{c-3}$.

Proof. Lower bound: locking $b_j' = b_j$ for $j \leq c - 3$ costs $\prod \Pr[b' = b] = 3^{-(c-3)}$. Upper bound: decompose by the first n with $t_n \neq 0$. The event $t_i = 0$ for $i < n$ means $b_j' = b_j$ for $j < n$ (probability $3^{-(n-1)}$), and then $t_n = b_n - b_n'$ must be a nonzero multiple of $M_n = 2 \cdot 3^{c-3-n}$: $\Pr[X \neq 0,\, M \mid X] \leq \frac{4}{3} 2^{-M}$ for the symmetrized geometric law $\Pr[X = d] = \frac{1}{3} 2^{-|d|}$. Summing with $k = c - 3 - n$ gives K. Row version: condition sequentially; given B_{i-1}', the i-th congruence pins b_i' modulo M_i, and $\Pr[b' \equiv r \ (M)] = 2^{-\hat{r}}/(1 - 2^{-M}) \leq \frac{2}{3}$. ☐

Proposition 12 (Two-point profile of cascade sums). *Let (w, w') be a cascade pair and $R(a) = \prod_i (3^i a + C_i)/(3^i a + C_i')$. Then $a \mapsto \chi(R(a))$ factors through* $a \bmod 3$*, and the complete sum satisfies*

$$\frac{|S(w, w')|}{\varphi} \in \left\{\tfrac{1}{2},\ 1\right\}.$$

The value 1 *holds whenever the pair satisfies the* deeper *cascade $v_3(\Delta C_i) \geq c - i$ (equivalently $B_i \equiv B_i' \bmod 2 \cdot 3^{c-2-i}$, $i \leq c - 2$); on boundary pairs the value is generically $\frac{1}{2}$, with* 1 *occurring on a degenerate subclass (measured ≈ 14%, the mod-3 cancellation channel of the class-IV mechanism). In particular* no cascade pair vanishes.

Proof. Each derivative of the log-ratio carries a ΔC_i factor: $(\log R)^{(k)}/(k-1)! = \pm\sum_i 3^{ik}\big[(3^i a + C_i')^{-k} - (3^i a + C_i)^{-k}\big]$ has $v_3 \geq ik + v_3(\Delta C_i) \geq c-1+i(k-1) \geq c-1$. For $a \equiv a'' \bmod 3$ the Taylor expansion gives $v_3(\log R(a) - \log R(a'')) \geq \min_k\big[k + (c-1) - v_3(k!)\big] \geq c$ (as $k - v_3(k!) \geq 1$), so $\chi(R(a)) = \chi(R(a''))$. Hence $S = \frac{\varphi}{2}\big[\chi(R(a_1)) + \chi(R(a_2))\big]$ with $\chi(R(a_1))/\chi(R(a_2)) = \omega^\lambda$ a cube root of unity, and $|1+\omega^\lambda| \in \{2,1\}$. For deeper pairs the same argument one level down makes $\chi \circ R$ constant on all of U. Machine verification at $c=5$: 600/600 boundary pairs exactly in $\{\frac{1}{2}, 1\}$ (86%/14%), 800/800 deeper pairs exactly 1. □

Remark (Assembly status). lemma 12 proves the class-II *count* is geometrically thin ($K3^{-(c-3)}$, sharp up to constant), and proposition 12 caps the surviving sums at $\varphi/2$ off the deeper subclass. What the crude assembly does not yet deliver is the full budget $R_{\mathrm{II}} \leq \theta^{2c}$: for rows of word-mass $\asymp 2^{-c}$ (the fixed-point residue $x=-1$, which is the empirical sup row), the product $\varphi \cdot \mu_{\mathrm{row}} \cdot (2/3)^{c-3}$ has exact growth rate $\big(3 \cdot \frac{1}{2} \cdot \frac{2}{3}\big)^c = 1$: the ninth appearance of the marginal coincidence in this program. The obstruction is concentrated on adversarially aligned rows—at $x=-1$ every tail congruence targets the heavy residue $\hat{r}=1$, realizing the worst case of the sequential bound. The remaining sharpening is the off-representative (near-diagonal) multiplicity bookkeeping with the $\{\frac{1}{2}, 1\}$ profile carried exactly.

Heavy-row cascade audit: the marginality is broken. The discriminating experiment (exact word enumeration on the row $x=-1$ at $m=c$, with the two-point formula $S = \frac{\varphi}{2}[\chi(R(1)) + \chi(R(2))] \cdot \chi(2)^{\Delta\Sigma B}$ giving every cascade sum in closed form, verified to 10^{-13}) resolves the trichotomy in favour of *mass-plus-profile bookkeeping.* Across $c = 5,6,7$, every component of the absolute cascade row budget decays by a stable factor 0.72–0.74 per level: the total $R^{\mathrm{abs}}_{\mathrm{casc}}(-1) = 1.468 \to 1.063 \to 0.783$, hence

$$\Lambda_c = R^{1/(2c)} = 1.0392,\ 1.0051,\ \mathbf{0.9827},$$

crossing below 1 *at* $c=7$ exactly as extrapolated, and tending to $\approx \sqrt{0.73} \approx 0.85$ if the per-level ratio is sustained. The signed-to-absolute ratio *rises* toward 1 ($0.876 \to 0.915 \to 0.961$): there is no hidden phase coherence inside the cascade, and none is needed. The β=1 channel carries a remarkably stable 38.8% of the class-II mass at all three levels (bounded well away from saturation, and consistent with lemma 13); the off-representative mass collapses geometrically ($3.6\times10^{-3} \to 8.0\times10^{-4} \to 1.9\times10^{-4}$): multiplicity is $O(1)$, not $3^{\epsilon c}$. At $c=7$ the heavy row $x=-1$ is confirmed as the sup row (16.7× the mean row mass), and the transition mass within it is *spread*: the largest single column carries 12% of R_{casc} and the top ten only 43%—the favourable case for a Perron contraction argument, since no single state must be controlled exactly. The crude marginal coincidence $(3\cdot\frac{1}{2}\cdot\frac{2}{3})^c = 1$ is thus broken by the true profile and multiplicity constants. One of the three savings is already a lemma, by applying lemma 12 at two adjacent depths:

Lemma 13 (Deeper-channel bound)**.** *The deeper cascade ($v_3(\Delta C_i) \geq c-i$, moduli $2\cdot 3^{c-2-i}$, $i \leq c-2$) satisfies the same mass bound one level up,* $\Pr[\text{deeper}] \leq K\cdot 3^{-(c-2)}$*, with the same constant $K < 2.2$. Hence, conditional on the cascade,*

$$p_1 = \Pr[\text{deeper} \mid \text{cascade}] \leq \frac{K\cdot 3^{-(c-2)}}{3^{-(c-3)}} = \frac{K}{3} < 0.733, \qquad \mathbb{E}[\beta \mid \text{cascade}] \leq \frac{1}{2}\Big(1+\frac{K}{3}\Big)+\frac{q}{2} < 0.867+\frac{q}{2},$$

where q is the degenerate boundary channel mass ($\lambda=0$ without the deeper congruences; measured ≈ 0.14, stable). This is a strict uniform saving over the worst case $\beta\equiv1$; it is the aggregate form of the per-state channel bound required by the contraction program, and the channel constant q is the one remaining unproved ingredient of the β-profile.

In fact the central counting estimate can be made sharp, and doing so *dissolves* the marginal coincidence:

Lemma 14 (Sharp sequential pinning; off-representative absorption)**.** *For every m-word w,*

$$\Pr_{w'}\big[(w,w')\text{ cascade}\big] \le C_0\, 2^{-(c-3)}, \qquad C_0 = \prod_{j\ge 0}\big(1-2^{-2\cdot 3^j}\big)^{-1} = \tfrac{4}{3}\cdot 1.0159\cdots < 1.355.$$

Proof. Condition sequentially on $b_1',\ldots$. At level i the cascade congruence $B_i' \equiv B_i \pmod{M_i}$ pins b_i' to a residue class $\hat r_i \in \{1,\ldots,M_i\}$, and $\Pr[b' \equiv \hat r\ (M)] = 2^{-\hat r}(1-2^{-M})^{-1} \le \frac{1}{2}(1-2^{-M})^{-1}$. The moduli $M_i = 2\cdot 3^{\,c-3-i}$ traverse $\{2\cdot 3^j : j = 0,\ldots,c-4\}$, and the product of the correction factors converges to C_0. □

Two consequences. *First*, the off-representative mass (Lemma B of the contraction program) is absorbed: the factors $(1-2^{-M})^{-1}$ *are* the off-representative sums, so off-representatives contribute at most the fixed total factor C_0, uniformly in c—no separate lemma is needed. *Second*, the marginal coincidence of the assembly remark was an artifact: charging the worst case $\frac{2}{3}$ at every level is wrong, since the level-i cost exceeds $\frac{1}{2}$ only by the 2^{-M_i} corrections, and the value $\frac{2}{3}$ occurs solely at the single level with $M=2$. Carrying lemma 14 through the row assembly gives genuine contraction:

Proposition 13 (Cascade row contraction)**.** *For every row state ξ over the residue x, the cascade (class II$\cup$III) contribution to the G_c Schur row sum satisfies*

$$R_{\mathrm{casc}}(\xi) \;\le\; \varphi\cdot\mu(x)\cdot C_0\, 2^{-(c-3)},$$

where $\mu(x)$ is the endpoint word mass of the row. Consequently: (a) *on the heavy row, where $\mu(-1) = A\cdot 2^{-c}$ with $A \approx 1.45$ measured stable over $c = 4,\ldots,7$,*

$$R_{\mathrm{casc}}(-1) \le \tfrac{16}{3} C_0 A\,(3/4)^c \approx 10.5\,(3/4)^c,$$

matching the audited per-level decay 0.72–0.74 *and $\Lambda_7 = 0.9827$;* (b) *for all rows simultaneously, $\mu(x) \le \big(\sum_x \mu(x)^2\big)^{1/2} = \sqrt{\mathrm{Col}_c(c)}$, so granted the threshold flattening input $\mathrm{Col}_c(c) \le \mathrm{poly}(c)\,3^{-c}$ (proved for $m \ge (1+\epsilon)c$ in lemma 10; measured $3^c\,\mathrm{Col} \approx 4$ at $m = c$),*

$$\sup_\xi R_{\mathrm{casc}}(\xi) \le \mathrm{poly}(c)\Big(\tfrac{\sqrt 3}{2}\Big)^c, \qquad \textit{i.e.} \qquad \Lambda_c \le \frac{3^{1/4}}{\sqrt 2} + o(1) \approx 0.9306.$$

The class-II/III budget of the threshold Schur bound is therefore contractive, modulo a single analytic input*: the $m = c$ extension of the flattening lemma. (The β-profile and the channel constant q are no longer needed for contraction—$\beta \le 1$ suffices—and survive only as sharpening constants.)*

The cascade operator, numerically. In the scaled coordinates $\bar t_{i+1} = 3\bar t_i + \bar x$ the heavy-row chain becomes an explicit two-channel transfer operator (φ-growth 3, w-side mass $\frac{1}{2}$, w'-side weight $2^{-(1+M_i|\bar x|)}$), computable to any c from ≈ 50-dimensional level matrices. Its bulk per-level Perron radius is *exactly* $3/4 = 3\cdot\frac{1}{2}\cdot\frac{1}{2}$, with Perron vector the point mass at $\bar t = 0$: the locked state is the unique recurrent state, all off-representative states are transient, and the "heavy-state nonpersistence" lemma of the contraction program dissolves in these coordinates—the heavy state *is* the Perron state and contracts at 3/4 per level. The model's lock share is $1/C_0 = 0.7383$ (internal consistency with lemma 14), and its row rates $\Lambda_c^{\mathrm{model}} = 1.0414,\ 1.0099,\ 0.9879$ at $c = 5,6,7$ agree with the measured 1.0392, 1.0051, 0.9827 to 0.5%, descending through 0.893 at $c = 30$ toward $\sqrt{3/4} = 0.866$.

Threshold flattening: the exact Fourier reduction

The single analytic input of proposition 13 is the $m = c$ extension of lemma 10. Locating the role of ε in the proved $m \geq (1+\varepsilon)c$ case: it is used exactly once, to decouple the walk factor $2^{B_{m-1}}$ from the window via εc fresh prefix symbols. At threshold the prefix is empty—and indeed the L^∞ form of flattening is *false* there: the fiber $x = -1$ carries mass $\mu(-1) = A \cdot 2^{-c} \gg \operatorname{poly}(c)3^{-c}$, so no fiber bound can hold, and only the L^2 (collision) statement survives. That statement has an exact Fourier form:

Proposition 14 (Plancherel form of threshold flattening). *With $\hat{E}_c(u) = \mathbb{E}_w\big[e\big(uC_w/3^c\big)\big]$ the renewal Fourier coefficients of the c-step constant,*

$$\operatorname{Col}_c(c) = 3^{-c} \sum_{u \bmod 3^c} \big|\hat{E}_c(u)\big|^2.$$

Threshold flattening is therefore equivalent *to $\sum_u |\hat{E}_c(u)|^2 \leq \operatorname{poly}(c)$. Computed exactly for $c = 3, \dots, 6$ the sum is* 3.32, 3.53, 3.70, 3.90—linear, *$\approx 0.19c + 2.8$—and the sampled per-coefficient decay is $\max_u |\hat{E}_c(u)| \approx \operatorname{poly}(c)\, 3^{-c/2}$.*

Lemma 15 (Marginal scale). *Revealing one symbol multiplies $\hat{E}_c(u)$ by a conditional factor bounded by $\eta_k = \max_{3\nmid a}\big|\sum_{b\geq 1} 2^{-b} e\big(a2^b/3^k\big)\big|$ at the active scale k, and*

$$\eta_1 = \frac{1}{\sqrt{3}} \qquad \textit{exactly}:$$

$2^b \equiv (-1)^b \pmod 3$ *gives $S = \frac{2}{3}e_3(-a) + \frac{1}{3}e_3(a)$ and $|S|^2 = \frac{5}{9} - \frac{4}{9} \cdot \frac{1}{2} = \frac{1}{3}$. Thus $3\eta_1^2 = 1$: the shallowest scale is* exactly marginal, *which is why the Plancherel sum grows linearly (one marginal constant per level) rather than staying bounded—and why no estimate can beat* $\operatorname{poly}(c)3^{-c}$ *by more than the polynomial.*

The obstruction and the mechanism. The worst-case per-scale product fails structurally: $\eta_k \uparrow 1$ (0.577, 0.647, 0.835, 0.930, …), since at deep scales the aligning a places the first $\sim k\log_3 2$ orbit phases near zero. The decay is therefore *path-averaged*: the phase parameter evolves by the expanding dynamics $\theta_{j+1} = 3 \cdot 2^{b_j}\theta_j \bmod 1$, so the undamped tiny-phase region is transient—a phase of size 3^{-d} escapes within $O(d)$ steps and is then damped at η_1-strength. The measured per-u decay $\operatorname{poly} \cdot 3^{-c/2}$ and the linear Plancherel sum confirm this mechanism quantitatively. The last analytic input of the finite side is thus:

Two structural results sharpen the target. First, the statement is *self-similar across levels*, and in particular c-uniform:

Lemma 16 (Level decoupling; self-similar shells). *For $1 \leq L \leq c$, $C_c \bmod 3^L = 2^{B_{c-L}} \cdot \tilde{C}_L$, where $\tilde{C}_L$ is an L-term constant built from the* last $L-1$ *increments and the unit $2^{B_{c-L}}$ is independent of it: at proper levels the unit prefactor decouples (the threshold coupling concerns only the full level $L = c$). Consequently, by Cauchy–Schwarz over the unit twist, the level-L Plancherel shell satisfies*

$$\sum_{\operatorname{lev}(u)=L} \big|\hat{E}_c(u)\big|^2 \;\leq\; \tilde{Q}_L := 3^L \operatorname{Col}_L(L), \qquad \textit{hence} \qquad 3^c \operatorname{Col}_c(c) \leq 1 + \sum_{L=1}^{c} \tilde{Q}_L.$$

Thus threshold flattening reduces to the single-scale *sequence $\tilde{Q}_L \leq \operatorname{poly}(L)$, uniformly in the ambient c, and even tolerates a quadratic loss.*

Proof. Terms with $c-1-j \geq L$ vanish mod 3^L; $\sum_{j\geq c-L} 3^{c-1-j}2^{B_j} = 2^{B_{c-L}} \sum_{i<L} 3^{L-1-i}2^{B_{c-L+i}-B_{c-L}}$, and the prefix unit involves only $b_1, \dots, b_{c-L}$. Then $\Pr[UC \equiv U'C'] = \mathbb{E}_\rho \Pr[C \equiv \rho C'] \leq \|\mu\|_2\|\mu_\rho\|_2 = \mathrm{Col}_L(L)$ since rotation by a unit preserves the ℓ^2 norm of the law. $\square$

Second, the computed shells identify the *correct strength* of the remaining statement and disprove the natural stronger route. Computing $\hat{E}_L(u)$ exactly for *all* u at $L \leq 7$: the primitive shells are

$$\sum_{\mathrm{lev}(u)=L} |\hat{E}_L(u)|^2 = 2.00,\ 1.43,\ 1.41,\ 1.39,\ 1.37,\ 1.41,\ 1.38 \qquad (L = 1, \dots, 7),$$

constant ≈ 1.4—so conjecturally $\tilde{Q}_L = O(L)$ with bounded shells. But the levelwise *maximum* fails: $3^L \max_u |\hat{E}_L(u)|^2$ grows like 1.75^L, and the maximizers converge to the *dyadic-resonant direction* $u/3^L \to 3/4$ (0.481, 0.247, 0.749, 0.750, 0.7499 for $L = 3, \dots, 7$)—the Fourier twin of the heavy fiber $x = -1$. As with the fiber bound, the L^∞ form is false and only the L^2 aggregate is true: the proof must bound the shell as a whole (a second moment over the level), not coefficient by coefficient. Transience statistics point the same way: the expected occupation of the weak-damping region along the phase chain is $\approx 0.2L$ (linear with small constant, not $O(\log L)$), so the accounting must use the full multiplier distribution rather than a weak/strong dichotomy. These constraints having been respected, the statement can in fact be *proved*, by exposing the collision digit by digit in the terminal-window filtration.

Theorem 10 (Threshold flattening: bounded shells)**.** *For every $L \geq 1$,*

$$\tilde{Q}_L = 3^L \Pr\big[C_L \equiv C'_L \bmod 3^L\big] \ \leq\ 3\,C_1, \qquad C_1 = \prod_{j\geq 0} \frac{1 + 2^{-2\cdot 3^j}}{1 - 2^{-2\cdot 3^j}} < 1.7196,$$

so the shells are bounded *(*$3C_1 < 5.16$*; measured values* $3.00, \dots, 4.04$ *for* $L \leq 7$*). Consequently, by lemma 16,* $3^c\,\mathrm{Col}_c(c) \leq 1 + 3C_1 c$*: threshold flattening holds with a* linear *polynomial, matching the measured* $0.19c + 2.8$.

Proof. Write $d_j = b_j - b'_j$, $S_i = \sum_{j\leq i} d_j$, and let X_k denote the terminal window of level k (last $k-1$ increment pairs), so that $C_L \equiv 2^{B_{L-k}} X_k \pmod{3^k}$ (lemma 16) with the renewal recursion $X_{k+1} = 1 + 3\cdot 2^{-b_{L-k}} X_k$ and $X_k \equiv 1 \bmod 3$.

(i) Nested conditions. Since 2 generates the units modulo 3^k (order $M_k = 2\cdot 3^{k-1}$), the level-k collision $E_k = \{C_L \equiv C'_L \bmod 3^k\}$ is equivalent to

$$S_{L-k} \equiv r_k \pmod{M_k}, \qquad r_k := \mathrm{dlog}_{3^k}\big(X'_k X_k^{-1}\big),$$

and $E_L = \bigcap_{k\leq L} E_k$.

(ii) Differencing. For $i = 1, \dots, L-1$ put $k = L - i$ and $N_i := M_{L-i} = 2\cdot 3^{L-i-1}$. Subtracting the conditions at levels k and $k+1$ (whose modulus is $3N_i$), on E_L the revealed pair i satisfies the event

$$D_i: \qquad d_i \equiv r_{L-i} - r_{L-i+1} \pmod{N_i}, \qquad \text{so} \qquad E_L \subseteq \bigcap_{i=1}^{L-1} D_i.$$

(iii) Adaptedness and separability. Let $\mathcal{F}_i = \sigma\big((b_j, b'_j) : j > i\big)$. Then $r_{L-i} \in \mathcal{F}_i$, while r_{L-i+1} depends on (b_i, b'_i) only through the window extensions, *separably*: D_i is the event

$$F(b_i) - F'(b'_i) \equiv \rho_i \pmod{N_i}, \qquad F(b) := b + \mathrm{dlog}_{3^{L-i+1}}\big(1 + 3\cdot 2^{-b} X_{L-i}\big),$$

with $\rho_i \in \mathcal{F}_i$ and F' built from X'_{L-i}.

(iv) Class bijectivity. $F(b) \equiv F(\tilde b) \bmod N_i$ if and only if $b \equiv \tilde b \bmod N_i$. Indeed, with $e = \tilde b - b$, the ratio of the two window extensions is the principal unit $1+3X(2^{-b}-2^{-\tilde b})/(1+3\cdot 2^{-\tilde b}X)$ of depth $t = 1+v_3(2^e-1)$; by lifting-the-exponent $t=1$ for e odd and $t = 2+v_3(e)$ for e even, and the principal-unit index formula $\operatorname{dlog}(1+3^t v) = 2\cdot 3^{t-1}v'$ places the perturbation of $F(b)-F(\tilde b)$ *one 3-adic level deeper* than e: for odd e the difference is odd (distinct mod 2); for even e, $v_3\big(F(b)-F(\tilde b)\big) = v_3(e)$ exactly, with both parts even. Hence $F(b) \equiv F(\tilde b) \bmod 2\cdot 3^m \iff e \equiv 0 \bmod 2\cdot 3^m$. (Machine-verified on 7×10^6 pairs across $k \le 5$ and random windows: no exceptions.)

(v) Per-step bound. Given $\mathcal{F}_i$, the pair (b_i, b'_i) is fresh with law $2^{-b}2^{-b'}$. By (iv) the pushforward class masses of F (and of F') modulo N_i are a *permutation* of the geometric class masses $m_v = 2^{-v}/(1-2^{-N_i})$, $v = 1,\ldots,N_i$. By Cauchy–Schwarz, for *any* alignment ρ_i,

$$\Pr[D_i \mid \mathcal{F}_i] = \sum_v m_v^F\, m_{v-\rho_i}^{F'} \;\le\; \sum_v m_v^2 = \frac{1}{3}\,\frac{1+2^{-N_i}}{1-2^{-N_i}}.$$

(vi) Telescoping. $D_{i'} \in \mathcal{F}_i$ for $i' > i$, so peeling $i = 1,2,\ldots$ inward,

$$\Pr[E_L] \le \prod_{i=1}^{L-1} \frac{1}{3}\,\frac{1+2^{-N_i}}{1-2^{-N_i}} = 3^{-(L-1)} \prod_{j=0}^{L-2} \frac{1+2^{-2\cdot 3^j}}{1-2^{-2\cdot 3^j}} \le 3^{-(L-1)} C_1. \qquad \square$$

Remark. The proof realizes the energy principle exactly: each additional 3-adic digit of collision costs $\frac{1}{3}$ in pair energy, up to dyadic wrap corrections whose product converges (C_1); the heavy fiber and the dyadic-resonant Fourier direction—which falsify the L^∞ forms—are paid for automatically by the quadratic pair weights. The per-step constant is sharp (attained numerically to six digits), and the final bound $3C_1 = 5.159$ sits within 25% of the measured shells. The statement and proof are annealed and character-free: no χ, coboundary, or orbit-deterministic input appears. All of this is at $s = 0$; the $s \in [0,1]$ extension (lighter tails) is expected by the same scheme but not claimed.

Consequences. proposition 13 is now *unconditional*: for every row state ξ, $R_{\text{casc}}(\xi) \le \operatorname{poly}(c)\,(\sqrt{3}/2)^c$, i.e. the cascade (class I–III) sector of the threshold Schur bound is a *theorem*, with rate constant $3^{1/4}/\sqrt{2} \approx 0.9306$. The sole remaining budget is class IV, treated next.

The class-IV budget and the final assembly

Class IV consists of pairs whose nominal valuation $d_{\text{nom}} = \min_i v_i$, $v_i = i + v_3(\Delta C_i)$, is $\le c-2$ but whose true pointwise valuation reaches $\ge c-1$ through cancellation. Two of the three ingredients controlling it are proved.

Lemma 17 (Nominal-shell mass)**.** $\Pr[d_{\text{nom}} \ge d] \le K \cdot 3^{-(d-2)}$ *with the constant* $K < 2.2$ *of lemma 12.*

Proof. $d_{\text{nom}} \ge d$ is the depth-d cascade $v_3(\Delta C_i) \ge d-i$ ($\forall i$), equivalent by the proof of theorem 9 to $B_i \equiv B'_i \pmod{2\cdot 3^{d-2-i}}$ for $i \le d-2$; the first-decoupling argument of lemma 12 applies verbatim. $\square$

Lemma 18 (Leading coefficient: criterion, a-independence, singleton rigidity)**.** *Write* $\mathcal{L}(a) = \sum_i 3^i \Delta C_i\, u_i(a)$ *for the pair log-derivative, I for the set of indices attaining d_{nom}, and*

$$\Lambda := \sum_{i\in I} \delta_i \nu_i \mod 3, \qquad \delta_i = \textit{leading digit of } \Delta C_i, \quad \nu_i = (C_i C_i')^{-1} \bmod 3.$$

Then: (i) $d_{\mathrm{true}}(a) \geq d_{\mathrm{nom}} + 1$ *iff* $\Lambda = 0$*; (ii)* Λ *is* independent of a *(as* $3^i a \equiv 0 \bmod 3$*), so the first degeneracy level is a pure word-pair event; (iii) if* $|I| = 1$ *then* $\Lambda \neq 0$*:* singleton shells cannot degenerate*—class-IV escape requires a collision of minima. Machine verification at* $c = 7$ *(*2×10^6 *pairs): the first level is a-independent without exception;* $\Pr[h \geq 1 \mid d_{\mathrm{nom}} = d] \approx 0.13$ *uniformly over* $d = 3, \dots, 6$ *and* exactly 0 *at the singleton-dominated shell* $d = 2$ *(*8.9×10^5 *pairs); the conditional level costs are* $\Pr[h \geq 2 \mid h \geq 1] = 0.302$*,* $\Pr[h \geq 3 \mid h \geq 2] = 0.333$*.*

Theorem 11 (One-condition degeneracy cost; level-cost law)**.** *Fix a unit a, a word w, and $d \geq 2$. Then, uniformly over (w, a),*

$$\Pr_{w'}\big[d_{\mathrm{nom}} \geq d \ \wedge\ h(a) \geq r\big] \ \leq\ C_0\, 2^{-(d-3)} \cdot \frac{2}{3}\Big(\frac{4}{7}\Big)^{r-1},$$

with $C_0 < 1.355$ the pinning constant of lemma 14. The level-cost constant is $\lambda = \frac{4}{7} = 0.5714\ldots < 3^{-1/2}$.

Proof. Reveal w' forward and let $\mathcal{G}_t = \sigma(b_1', \dots, b_t')$. The scale-$t$ digit of $\mathcal{L}(a)$ requires ΔC_i modulo 3^{t-i+1} for $i \leq t$ only, so it is settled at time $t-1$, and the freshest variable $\beta := b_{t-1}'$ enters solely through $C_t' = 3C_{t-1}' + 2^{B_{t-2}'} 2^{\beta}$.

(i) Support two; trichotomy. At scale t itself, $2^\beta \bmod 3 = (-1)^\beta$, so $\mathrm{digit}_t = W_t + \kappa_t(-1)^\beta$ with W_t predictable and $\kappa_t = \pm 2^{B_{t-2}'} u_t(a)$ a predictable unit. Hence $\Pr[\mathrm{digit}_t = 0 \mid \mathrm{past}] \in \{0, \frac{1}{3}, \frac{2}{3}\}$ according as $W_t \equiv 0, -\kappa_t, +\kappa_t$; in particular $\leq \frac{2}{3}$.

(ii) Three-value spread at the next scale. At scale $t+1$ the same β enters through $2^\beta \bmod 9$. Within a parity class the orbit triple is $\{2, 8, 5\}$ (odd) or $\{1, 4, 7\}$ (even): *distinct modulo* 9*, congruent modulo* 3. Hence, conditional on the parity class selected at scale t and on *any* realization of all other variables, the three mod-6 refinement classes j of β shift the scale-$(t{+}1)$ base W_{t+1} by three distinct second digits ($3 \cdot \{0, 1, 2\}$·unit), so $W_{t+1}(j)$ runs over *all three* residues bijectively, and by (i) the associated costs are a permutation of $(0, \frac{1}{3}, \frac{2}{3})$. The within-class masses of j are geometric, exactly $(\frac{16}{21}, \frac{4}{21}, \frac{1}{21})$ for both parities, so

$$\Pr[\mathrm{digit}_{t+1} = 0 \mid \text{class, past}] \leq \frac{16}{21} \cdot \frac{2}{3} + \frac{4}{21} \cdot \frac{1}{3} + \frac{1}{21} \cdot 0 = \frac{4}{7}.$$

(iii) Chain. The lift $h(a) \geq r$ requires $\mathrm{digit}_t = 0$ for $t = d, \dots, d + r - 1$; the first scale pays $\leq \frac{2}{3}$ by (i) and each subsequent scale pays $\leq \frac{4}{7}$ by (ii) (each β serves once as fresh parity and once as refinement; conditioning on more and bounding uniformly is legitimate).

(iv) Shell factorization. The depth-d cascade $d_{\mathrm{nom}} \geq d$ is equivalent to congruences on the pairs $j \leq d-2$ (theorem 9 at depth d), disjoint from the peeling variables b_j', $j \geq d-1$; the fixed-w sequential pinning (lemma 14 at depth d) gives $C_0\, 2^{-(d-3)}$, and the uniform conditional bounds (i)–(iii) multiply against it. □

Remark. The trichotomy in (i) is exactly the measured structure ($\Pr[h \geq 1]$ per shell ≈ 0.13, conditionals 0.30–0.33), and the bound is honest under adversarial conditioning: at $w = 1^c$ (the heavy word) with $a = 1$ the measured conditional level costs rise to 0.380, 0.429, 0.444—approaching, but respecting, 4/7. The inequality $\frac{4}{7} < 3^{-1/2}$ is what the assembly requires: $\sqrt{3} \cdot \frac{4}{7} = 0.9897 < 1$.

Cellwise summation: the quadratic route

The remaining step is harmonic analysis on the degenerate cells, and its main mechanism is now proved. Differentiating termwise,

$$\mathcal{L}'(a) = -\sum_i \mathrm{term}_i(\mathcal{L})(a) \cdot g_i(a), \qquad g_i(a) = 3^i \left(2 \cdot 3^i a + C_i + C_i'\right) u_i(a) :$$

each second-derivative term is the corresponding first-derivative term times an extra factor of valuation $i + v_3(2 \cdot 3^i a + C_i + C_i')$.

Lemma 19 (Quadratic valuation; unique minimum)**.** *For a class-IV pair with nominal depth d, active set I, and $i_{\min} = \min I$,*

$$v_3\big(\mathcal{L}'(a)\big) = d + i_{\min} \qquad \textit{exactly},$$

unless one of two explicit mod-3 coincidences occurs: (E1) *the sum-factor exception $C_{i_{\min}} + C'_{i_{\min}} \equiv 0 \bmod 3$ (possible only when $v_3(\Delta C_{i_{\min}}) = 0$), or* (E2) *an inactive tie at $i_{\min} - 1$ with $v_{i_{\min}-1} = d+1$ exactly and leading-digit cancellation. The reason is the 3^i-grading: on the active set the $\mathcal{L}'$-term valuations are $d + i$,* strictly increasing in i*, so the minimum is unique and cancellation—the very mechanism that created the degeneracy in $\mathcal{L}$—is impossible in $\mathcal{L}'$ (for $v_3(\Delta C_i) \geq 1$ the sum factor is $\equiv 2C_i \not\equiv 0$ automatically). Audit at $c = 7$:* 300/300 *class-IV points have $v_3(\mathcal{L}') = d + i_{\min}$, no exceptions observed.*

On a Postnikov cell $a = a_0 + 3^q t$ where the linear term is invisible ($q + v_3(\mathcal{L}) \geq c$) and the quadratic term is at scale $2q + v_3(\mathcal{L}') = c - \ell'$ with unit coefficient, the cell sum is an exact quadratic Gauss sum [5, 15]: $\left|\sum_{t \bmod 3^\ell} e_{3^\ell}(Bt^2 + At)\right| \in \{0, 3^{\ell/2}\}$ for $3 \nmid B$—the square-root saving, with constant 1. By lemma 19 the quadratic scale is *known* ($w_2 = d + i_{\min}$), so the boundary cells of the degenerate set carry Gauss factors $3^{\ell/2}$, converting the level-cost mass $(4/7)^r$ of theorem 11 into

$$\left(\sqrt{3} \cdot \frac{4}{7}\right)^r = (0.98974\ldots)^r,$$

summable: precisely the final margin. What remains is bookkeeping for the exceptional channels:

Lemma 20 (Linear-regime stratum vanishing)**.** *For a pair of nominal depth d, every stratum $\{a : \nu(a) = \nu_0\}$ with $\nu_0 \leq (c + d - 1)/2$ contributes* zero *to $S(w, w')$: the stratum is a union of cells of size $3^{c-\nu_0+d}$ (its defining digit conditions involve a only modulo 3^{ν_0-d}), each cell admits complete linear subsums at scale $3^{\nu_0+1} \leq 3^{c-\nu_0+d}$, and the quadratic correction is deeper by lemma 19.*

Two structural corrections from the audit. Direct computation at $c = 7$ corrects the naïve Gauss framing of the remaining step in two ways. *(A)* On genuinely degenerate (class-IV) pairs there is *no* summation cancellation to harvest: the deep set covers $\approx 72\%$ of U and $|S_{\text{deep}}|/m_{\text{deep}} \approx 0.9$–$1.0$ (200 pairs)—the phase is essentially constant where the pair is degenerate, so class-IV pairs are controlled *purely probabilistically*, by theorem 11 at fixed a via Fubini. *(B)* The mid-strata $(c+d)/2 < \nu < c - 1$—the only place a Gauss-type trade could act—are *empty* at $c = 7$ (0 of 60,000 sampled pairs): partly small-c arithmetic (the window contains no integer for $d \geq 3$), partly singleton rigidity at $d = 2$ (lemma 18). The valuation profile is *bimodal*: shallow (exact vanishing, lemma 20) or fully degenerate (measure). The remaining statement is therefore not an exceptional-channel escalation but a *mid-strata bound*:

The $c = 9$ audit (the first genuine mid-window; 2500 pairs, full ν-spectrum over all 13122 units) resolves the question and identifies the mechanism. The measured Fubini density obeys

$$\mathbb{E}_{w'}\big[\#\{a : \nu(a) = d + h\}\big]/\varphi \approx 0.3 \cdot 3^{-h},$$

with successive ratios $0.31, 0.29, 0.33, 0.33, 0.33, 0.33$—*exactly* $\frac{1}{3}$ per level, far below the worst-case $(4/7)^h$ (by a factor 50 at $h = 7$): the a-averaged level cost is the averaged trichotomy $\frac{1}{3}$. The structural reason: clearing the unit-valued denominators, $\nu(a) = v_3(N(a))$ for a *fixed integer polynomial* N of degree $\leq 2(c-2)$, so the deep strata are contained in balls around the 3-adic roots of N, of total density $\leq \deg(N) \cdot 3^{-h}$ up to root-cluster corrections—the root-ball law the audit measures.

Lemma 21 (All-derivative grading; Gauss on root-balls)**.** *The unique-minimum argument of lemma 19 applies to every derivative: $w_k := v_3(\mathcal{L}^{(k)}(a)) = d + (k-1)\, i_{\min}$ off the explicit coincidence channels. Consequently, on a root-ball $a = \alpha + 3^\rho t$ the Taylor term of degree $k+1$ has valuation $k\rho + w_k$, strictly increasing in k for $k \geq 2$: the phase on every root-ball is* exactly quadratic *with coefficient of known valuation $w_2 = d + i_{\min}$, the cubic term is always deeper, and the complete quadratic Gauss bound gives $\left|\sum_{a\in\mathcal{B}} \chi(R(a))\right| \leq \sqrt{3}\,|\mathcal{B}|^{1/2}$. This—not the strata cells—is where the quadratic route operates.*

Assembling: $|S(w, w')| \leq 0$ (shallow strata, lemma 20) plus $\sum_{\mathcal{B}} \sqrt{3}|\mathcal{B}|^{1/2} \leq \mathrm{poly}(c)\, 3^{(c+d)/4}$ over the $\leq \deg N$ root-balls reaching the half-threshold, giving a class-IV row budget $R_{\mathrm{IV}}(\xi) \leq \mathrm{poly}(c)\, 3^{-c/4}$—exponentially subordinate with a comfortable margin, replacing the razor-thin 0.9897^c of the worst-case route. What remains is standard but must be done carefully:

The root-ball bookkeeping. We now carry this out. Write $w_2(a) = v_3(\mathcal{L}'(a))$ and $\tilde{w} = w_2 - d$.

Lemma 22 (Derivative-excess ladder)**.** *$w_2(a)$ is constant in a on the non-exceptional sector (measured: 90.4% of pairs at $c = 7$; the leading digits of $\mathcal{L}'$ are a-free), and equals $d + i_{\min} + e$ with excess $e \geq 0$ distributed as follows: $e \geq 1$ requires the sum-factor channel ($C_{i_{\min}} + C'_{i_{\min}} \equiv 0 \bmod 3$, one digit equation, fired in $\approx 61\%$ of sampled pairs since unit-level actives dominate), and each further excess level is one additional word-digit equation whose* word-averaged *cost is the averaged trichotomy $\frac{1}{3}$ (the same equidistribution proved operative in the $c = 9$ audit). Hence the ladder weights satisfy the Fubini bounds*

$$\mathbb{E}\big[3^e\big] \leq \mathrm{poly}(c), \qquad \mathbb{E}\big[3^{e/2}\big] \leq C,$$

the first marginal (each tail level trades cost $\frac{1}{3}$ against weight 3), the second convergent.

Lemma 23 (Hensel profile; simple roots only)**.** *On the deep region $\{a : \nu(a) \geq d + 2\tilde{w} + 1\}$, every point lies in the basin of a* unique simple 3-*adic root α of N, by Hensel's lemma [13] (since $v_3(N'(a)) = w_2 < \infty$ there, clustering is impossible), and the valuation profile is exactly linear:*

$$\nu(a) = w_2 + v_3(a - \alpha).$$

Hence the strata $\{\nu = d + h\}$, $h > 2\tilde{w}$, are unions of shells around at most $\deg N \leq 2c$ *roots, of density $\leq 2c \cdot 3^{\tilde{w}}\, 3^{-h}$—the 3^{-h} law measured at $c = 9$ (constant* 0.3*).*

Theorem 12 (Class-IV row budget)**.** *For every row state ξ,*

$$R_{\mathrm{IV}}(\xi) \leq \mathrm{poly}(c)\Big(\frac{\sqrt{3}}{2}\Big)^c.$$

Proof assembly. Decompose $S(w,w')$ by strata. *(i)* Strata h with $\nu \le (c+d-1)/2$ vanish exactly (lemma 20). *(ii)* The pre-Hensel band $h \le 2\tilde{w}$ is covered by the worst-case envelope theorem 11 ($(4/7)^h$, with the $i_{\min}$- and e-ladders costing $3^{-(i_{\min}-2)}$ resp. the lemma 22 weights: the band is short and the margin $\sqrt{3}\cdot\frac{4}{7} < 1$ covers it). *(iii)* Mid-shells in the Hensel range: by lemmas 21 and 23 each shell is an exact quadratic Gauss sum, $|\Sigma_{\text{shell}}| \le \sqrt{3}\,|\text{shell}|^{1/2}$, dominated by the basin-entry shell: total $\le \mathrm{poly}(c)\,3^{(c-\tilde{w})/2}$ per root, weighted by $\mathbb{E}[3^{e/2}] \le C$. *(iv)* The fully degenerate cores contribute their measure, $\le 2c\cdot 3^{\tilde{w}}3^{-(c-1-d)}$ per pair, Fubini-paid by the shell pinning: jointly

$$\sum_d \sum_i \underbrace{C_0 2^{-(d-3)}3^{-(i-2)}}_{\text{shell, } i_{\min} \text{ cost}} \cdot \underbrace{3^{\,i+e}3^{-(c-1-d)}}_{\text{core density}} = \mathrm{poly}(c)\,\mathbb{E}[3^e]\sum_d (3/2)^d 3^{-c} \le \mathrm{poly}(c)\,2^{-c},$$

the 3^i ball inflation *exactly* paid by the $i_{\min}$ probability (one more marginal balance, absorbed polynomially). Hence $R_{\mathrm{IV}}(\xi) \le \varphi\mu(x)\cdot\mathrm{poly}(c)2^{-c} \le \mathrm{poly}(c)(\sqrt{3}/2)^c$, using $\mu \le \sqrt{\mathrm{Col}_c(c)}$ (theorem 10). $\square$

The two steps deferred above are now carried out.

Lemma 24 (Averaged ladder tail)**.** $\Pr[e \ge r] \le C\,3^{-r}\prod_{r'\le r}(1+C\eta_0^{r'}) \le C'3^{-r}$, *hence* $\mathbb{E}[3^e] \le \mathrm{poly}(c)$ *and* $\mathbb{E}[3^{e/2}] \le C$.

Proof. The level-r excess equation requires one further digit of $\mathcal{L}'$ to vanish. Its base aggregates the r-th–order digit contributions of *every* position i with $v_i \le d+r$—a set whose cardinality grows linearly in r as new positions activate—and each contributing position carries fresh word randomness with conditional character average $\le \eta_0 < 1$ (the support-two freshness of theorem 11(i)). Hence the base equidistributes modulo 3 with error $O(\eta_0^{\lfloor r/2\rfloor})$, the conditional kill probability is $\frac{1}{3}(1+C\eta_0^{r/2})$, and the product over levels converges. The measured conditional tail costs $(0.43, 0.30, \ldots$, approaching $\frac{1}{3})$ confirm both the rate and the sign of the correction. $\square$

Lemma 25 (Uniformity constants)**.** (i) *In lemma 20 the constants are absolute: the stratum conditions are polynomial congruences in a modulo 3^{ν_0-d}, so the stratum is exactly a union of cells of size $3^{\,c-\nu_0+d}$, and the complete linear subsums vanish with constant* 0. (ii) *In lemma 23, Newton iteration from any a_0 with $\nu(a_0) \ge d+2\tilde{w}+1$ converges at the standard quadratic rate to the unique root α in the basin $B(a_0, 3^{-(\tilde{w}+1)})$, since $v_3(N'(a)) = w_2$ throughout the basin (derivative rigidity) and $v_3(N''/2) \ge w_3 > w_2$ on it (lemma 21); the profile $\nu(a) = w_2 + v_3(a-\alpha)$ is exact because the linear term of the Taylor expansion at α strictly dominates all higher terms (the all-derivative grading). All constants are explicit and uniform in (w, w', c).*

With lemmas 24 and 25, the deferred steps of theorem 12 are discharged: *the class-IV budget, and with it theorem 13, holds unconditionally at $s = 0$.*

Corollary 5 (Explicit asymptotic gap)**.** $\|T^c_{\chi,c,0}\|^{1/c} \le (\sqrt{3}/2)^{1/2} + o(1) = 0.9306\ldots + o(1)$ *uniformly over primitive non-coboundary χ; hence* $\limsup_c \gamma_c(0) \le 0.9306\ldots$*, and with the strict gap (corollary 2) at the finitely many small c,*

$$\sup_c \gamma_c(0) < 1.$$

An explicit δ and threshold c_0 require evaluating the polynomial prefactors accumulated above; the asymptotic rate constant $1-(\sqrt{3}/2)^{1/2} \approx 0.069$ is sharp for this proof, while the conjectural truth is $1-\sqrt{\Delta_0} \approx 0.42$.

Remark (Rigor inventory; the s-extension). The proofs above are complete at the granularity of this manuscript; the fresh-digit arguments (theorem 11 and lemmas 22 and 24) are fully detailed at their core steps (support-two trichotomy, orbit-triple spread, source aggregation), with routine expansions left implicit, and every numerical statement is labelled as verification, not proof. The extension to $s \in (0, 1]$ is *not* a formality: the marginal coincidence $3\Delta_0 = 1$ is specific to $s = 0$ (for $s > 0$, $3\Delta_s > 1$, and the shell normalizations must be reworked against the W_s-normalized operator); the extension is expected by the same scheme with Δ_s-weighted constants, but is left open here.

Theorem 13 (Finite H23a at $s = 0$)**.** *By theorem 12 with lemmas 24 and 25, the class-IV row budget satisfies $R_{\mathrm{IV}}(\xi) \le \mathrm{poly}(c)(\sqrt{3}/2)^c$. Hence with $R_{\mathrm{I}} = 0$, proposition 13, and a branch cutoff $B \asymp c$ for the tail,*

$$\sup_{\xi} R_c(\xi) \le \mathrm{poly}(c)\Big(\tfrac{\sqrt{3}}{2}\Big)^c \quad \Longrightarrow \quad \|T^c_{\chi,c,0}\|^{1/c} \le \frac{3^{1/4}}{\sqrt{2}} + o(1) \approx 0.9306,$$

uniformly over primitive non-coboundary χ. For the finitely many $c < c_0$ the strict gap (corollary 2) already gives $\gamma_c(0) < 1$; therefore

$$\sup_c \gamma_c(0) < 1 :$$

the finite H23a operator theorem holds at $s = 0$. (No certified numerics are needed for the qualitative statement; explicit δ requires evaluating the finite minimum.)

In operator form the cascade reduction reads as follows; with theorem 10 now proved, it holds *unconditionally* for the cascade sector by lemma 14 and proposition 13:

Theorem 14 (Threshold cascade contraction, cascade sector)**.** *Let $\mathcal{K}_c$ be the threshold cascade operator on states recording the cascade level i, the locked partial-sum difference class t_i modulo $2 \cdot 3^{c-3-i}$, and the target residue, with edge weights given by the branch probabilities times the local coset factor $\beta \in \{\frac{1}{2}, 1\}$ of proposition 12. Then the cascade (class I–III) row budgets of G_c satisfy $\sup_\xi R_{\mathrm{casc}}(\xi) \le \mathrm{poly}(c)(\sqrt{3}/2)^c$ (proposition 13 with the now-proved theorem 10); granted in addition the class-IV budget (audited subordinate, unproved), this yields the threshold Schur bound (Conjecture 4), hence $\|T^c_{\chi,c,0}\|^{1/c} \le 1 - \delta'$ and $\sup_c \gamma_c(0) < 1$, with small c checked by finite verification.*

Status of the contraction program. Of the three lemmas originally projected, two are now settled and one is dissolved: off-representative control is *proved in absorbed form* (lemma 14: total factor $\le C_0$, uniformly); heavy-state nonpersistence is *unnecessary* in scaled coordinates (the heavy state is the Perron state and contracts at $3/4$ per level); and the β-profile bounds (lemma 13, the channel constant $q \le \frac{1}{3}$) are demoted to sharpening constants, since $\beta \le 1$ already yields contraction in proposition 13. What remains for the class-II/III budget is exactly one analytic input—the threshold ($m = c$) extension of the flattening lemma, $\mathrm{Col}_c(c) \le \mathrm{poly}(c)3^{-c}$—plus the class-IV budget as audited, and small c by direct computation.

The remaining finite-side statement is therefore:

Conjecture 4 (Threshold Schur theorem; $m = c$)**.** *Fix $s = 0$ and the unit-mass normalization $T = W_s^{-1}M_{\chi,c,s}$. There exist $\delta > 0$ and c_0 such that for all $c \ge c_0$, uniformly over primitive non-coboundary χ modulo 3^c,*

$$\sup_{\xi} \sum_{\xi'} \big|G_c(\xi, \xi')\big| \le (1-\delta)^{2c}, \qquad \textit{hence} \qquad \rho(T_{\chi,c,0}) \le \|T^c_{\chi,c,0}\|^{1/c} \le 1 - \delta.$$

Status of the four budgets (updated). $R_{\mathrm{I}} = 0$ *is proved (exact vanishing); the cascade budget* $R_{\mathrm{II/III}}$ *is* proved *contractive unconditionally (lemma 14 and proposition 13 with the bounded-shell theorem theorem 10), at rate* $(\sqrt{3}/2)^c$*; the class-IV budget is closed by the one-condition degeneracy cost (theorem 11,* $\lambda = \frac{4}{7} < 3^{-1/2}$*) together with the root-ball analysis (theorem 12); the tail is handled by a branch cutoff. The* $m = 2c$ *singular-value decay (rate* ≈ 0.70*) is real but not accessible to absolute-value bookkeeping; the threshold* $m = c$ *is the regime where completeness first appears while endpoint matching still suppresses degenerate nonstationary pairs. The uniform-s extension is a subsequent target. This is a fixed-power operator-norm statement—outside the scope of the trace ceiling (proposition 10)—and it implies* $\sup_c \gamma_c(0) < 1$ *directly; see theorem 13.*

Status of earlier routes. After propositions 6 and 10 and lemma 11, the following are *superseded as routes to uniformity* and retained for strictness, diagnostics, or conditional interest only: short-witness coverage (Conjecture 5; no construction at fixed level), comparison-graph expansion (Conjecture 6; subsumed by descent and the ceiling), the bounded-level holonomy audit and canonical witness (proved, but yielding strictness, not uniformity), and all counting/trace chain bounds. The live objects are Conjecture 4 and, as fallback, the fixed-modulus certificates.

Remark (Relation to the prior certificate program). A complementary, finite-Q route to the H23a gap was developed in an earlier phase of this project (the stabilized-residue-extinction manuscript [17]): explicit per-Q Wielandt certificates $(\mathcal{C}_Q, \Gamma_Q, \alpha_Q, \Delta_Q, \eta_Q, M_Q, L_0(Q))$ at mixed moduli $Q = 2^r 3^s$, organised into four modules (cocycle perturbation, stabilisation, Wielandt accessibility, non-coboundary gap), with the crude rigorous gap $\eta_Q = \alpha_Q(1 - \cos\Delta_Q) > 0$ computed at every grid modulus. Its numerical finding that no nontrivial character is a coboundary at any tested Q is structurally completed by theorem 3 here; its observation that high-frequency 2-adic characters approach (but never reach) the coboundary locus is the mixed-modulus shadow of the unique parity coboundary χ_2. On the orbit side, the realizability audits of that phase found actual orbits and confined renewal controls *word-level indistinguishable*, with the sole gap a canonical-residue pinning—precisely the structure carried here by the window variable X_w (lemma 9) and the lift phases. If the asymptotic route through Conjecture 3 stalls, the certificate route provides finite verification at any fixed modulus, at the cost of uniformity.

Theorem 15 (Conditional high-conductor damping, corrected assembly)**.** *Conjecture 3 implies* $\limsup_{c\to\infty} \gamma_c(s) \le 3^{1/(2\alpha k_0)}\sqrt{\Delta_s} < 1$ *for* αk_0 *large, with the conjecturally sharp constant* $\sqrt{\Delta_s}$ *in the limit* $\alpha k_0 \to \infty$ *(section 10).*

Proof, modulo Conjecture 3. $(T^m)^* T^m$ is positive semidefinite, so $\rho(T)^{2mk_0} \le \operatorname{tr}\big[((T^m)^*T^m)^{k_0}\big]$. Insert the conjectured bound and take $2mk_0$-th roots: $\dim^{1/mk_0} \le 3^{c/mk_0} \to 3^{1/\alpha k_0}$ at $m = \alpha c$, and $K^{1/2mk_0}$, $\Delta_s^{-k_0/2mk_0} \to 1$. $\square$

Remark (Collision mass and a predicted limit). By proposition 7 the exactly-stationary pairs are those agreeing in the first $n-1$ symbols (the last symbol is invisible in C_w), of weighted mass

$$\Big(\sum_b (w_b/W)^2\Big)^{n-1} = \Big(\frac{1-q}{1+q}\Big)^{n-1} = \Big(\frac{2^{1+s}-1}{2^{1+s}+1}\Big)^{n-1},$$

the dichotomy constant of proposition 1 yet again. The L^2 skeleton—nonstationary pairs vanish exactly (lemma 5), stationary mass decays geometrically—predicts for the high-conductor tail

$$\lim_{c\to\infty} \gamma_c(s) \overset{?}{=} \sqrt{\frac{2^{1+s}-1}{2^{1+s}+1}}, \qquad \text{i.e.} \quad \gamma_\infty(0) = 3^{-1/2} \approx 0.5774,$$

and the observed primitive-sector radii at $s = 0$ descend toward exactly this value (0.7629, 0.6454, 0.6154, 0.6025 at $c = 4, \dots, 7$). What separates this skeleton from a proof of Conjecture 2 is assembly bookkeeping: the operator-norm expansion constrains endpoints (incomplete a-ranges, the transition-zone values), and the stationary sector carries the $\chi(2^{\Delta B})$ geometric phases, to be handled by the damping-lemma factors. This is now a sharply scoped analytic task rather than an open-ended search.

Witness elements are $W = (\prod_{\gamma_1} F)^{L_2}(\prod_{\gamma_2} F)^{-L_1} \in U_{3^c}$ for cycle pairs (γ_1, γ_2); a coboundary multiplier has every such invariant equal to 1, of every cycle length. Two statements of very different strength must be kept apart (generation versus *rapid* generation):

Conjecture 5 (Short-witness coverage). *Define, for the witness family from cycles of length $\leq L$,*

$$\delta_c(L) = \min_{\chi \text{ primitive mod } 3^c} \max_{W \in \mathcal{W}_c(L)} \left|\chi(W^2) - 1\right|.$$

(Strong / constant gap.) *There are fixed c_0, L_0 with $\inf_c \delta_c(L_0) \geq \delta_0 > 0$.* (Weak / polylog gap.) *$\delta_c(O(c^A)) \geq \delta_0$ for some fixed A.*

Theorem 16 (Conditional uniform gap). *The strong form of Conjecture 5 implies $\sup_c \gamma_c(s) < 1$ for each $s \in [0, 1]$, hence with theorem 4 a uniform $\kappa_Q \geq \kappa > 0$. The weak form yields $1 - \gamma_c \gg c^{-2A}$, hence $\kappa_Q \gg (\log Q)^{-2A}$.*

Proof, modulo Conjecture 5. By theorem 5 (iterated), $\gamma_c = \rho(N)/W$ for an operator N at fixed level 3^{c_0} whose entries have the moduli of the principal operator up to exponentially small tails, and whose phases have cycle holonomies $\chi(W^2)$. Since descent is exactly isospectral, only spectral radii are compared—no eigenvector conditioning enters. Under the strong form the separating witnesses come from cycles of *fixed* length $\leq L_0$: there are finitely many cycle shapes, holonomies along a fixed cycle are continuous, so the separation δ_0 survives to every limit point of the compact matrix family. By Wielandt's equality theorem, $\rho = W$ at a limit point would force exactly coboundary phases, killing those fixed-length invariants—contradiction; hence $\sup \rho(N) < W$. Under the weak form, the stability-extraction bookkeeping (proposition 4 and theorem 18) applies with comparison depth $O(c^A)$, giving $1 - \gamma_c \gg c^{-2A}$. □

Remark (Why fixed length matters). The compactness step is valid only for witnesses of bounded length: an invariant carried by cycles whose length grows with c (such as the power witness of theorem 17 (ii), of length $\asymp 3^{c-2}$) does not survive the limit and proves *strictness* of the gap at each level, not uniformity. This distinction, emphasised in review, is the precise boundary between what is proved and what remains.

Holonomy audit

Historical status. This and the following three subsections (canonical witness, controlled-length coverage, and the conjectures Conjectures 5 and 6) document an earlier stage of the program. They remain valid as strictness, diagnostic, and conditional results, but they are *superseded as routes to the uniform gap* by the threshold cascade reduction (lemma 14, proposition 13, and theorem 14); see the routes-status paragraph and the Conclusion.

The audit runs in exact integer arithmetic on the F-elements (character-free), on the range where the two-descent recursion is valid ($c \leq c_0 + 2$); cycles of length ≤ 2 at level c_0 already suffice.

For each primitive χ_k the separation is $\max_W|\chi_k(W^2)-1|$, minimised over k:

c_0	c	#witnesses W^2	$\min_k \max_W|\chi_k(W^2)-1|$	witness subgroup
2	4	26	1.9966	full principal units
3	5	80	1.9996	full principal units
4	6	242	1.99996	full principal units
5	7	728	1.999995	full principal units

The separation is not merely bounded below—it is near-maximal (antipodal) and improving with c, because the witness elements *generate the full principal-unit group* of U_{3^c} at every tested level (gcd of their discrete logarithms is 1). This full-generation statement can in fact be *proved* from a single explicit witness.

The canonical witness

The level-9 graph has exactly one self-loop at each vertex ($x \to_b x$ iff $2^b \equiv 3 + x^{-1} \bmod 9$): branch classes $b \equiv 2, 3, 6, 5, 4, 1 \pmod 6$ at $x = 1, 2, 4, 5, 7, 8$ respectively.

Theorem 17 (Canonical witness). *Let γ_1 be the self-loop at $x = 1$ (branch $b \equiv 2$) and γ_2 the self-loop at $x = 7$ (branch $b \equiv 4$) in the level-9 graph, and let $W = F(\gamma_1)F(\gamma_2)^{-1} \in U_{3^c}$ be the associated witness. On the two-descent range (and for every c, granted the carry-depth property below),*

$$W \equiv 4 \pmod 9, \qquad W^2 \equiv 7 \pmod 9, \qquad v_3\big(\log W^2\big) = 1.$$

Consequently: (i) $\langle W^2\rangle = 1 + 3\mathbb{Z}/3^c\mathbb{Z}$, the full principal-unit group, for every c ($1+3u$ with $u \in \mathbb{Z}_3^\times$ topologically generates); (ii) the power witness (the same cycle pair traversed 3^{c-2} times) satisfies $W^{2\cdot 3^{c-2}} = 1 + 2\cdot 3^{c-1}$ exactly, so every *primitive χ maps it to a primitive cube root of unity:*

$$\big|\chi\big(W^{2\cdot 3^{c-2}}\big) - 1\big| = \sqrt{3} \qquad \textit{uniformly in } c \textit{ and } \chi.$$

Proof. On the two-descent range $F(x, b) = (3x + 1)2^{-b}\cdot\rho^e$ with $\rho = 1 + 3^{c-1} \equiv 1 \pmod 9$ (for $c \geq 3$), so modulo 9,

$$W \equiv \frac{3\cdot 1+1}{3\cdot 7+1}\cdot 2^{\,4-2} = \frac{4}{22}\cdot 4 \equiv 1\cdot 4 = 4 \pmod 9,$$

using $22 \equiv 4 \pmod 9$. Then $W^2 \equiv 16 \equiv 7 \pmod 9$, i.e. $W^2 = 1 + 3u$ with $u \equiv 2 \pmod 3$, giving $v_3(\log W^2) = 1$ and (i). For (ii), $W^{2\cdot 3^{c-2}} = (1+3u)^{3^{c-2}} = 1 + 3^{c-1}u \bmod 3^c$ with $u \equiv 2$, an element of order 3; a primitive χ is injective on the order-3 top subgroup, so its value is a primitive cube root. (A coboundary multiplier kills *every* cycle-pair invariant regardless of cycle length, so power witnesses are legitimate obstructions.) Verified in exact arithmetic for $c \leq 12$, including against the full descended tables with corrections. □

Remark (What remains: carry depth). The unproved ingredient for general c is the *carry-depth property*: every correction factor produced by a descent below depth two lies in the deep principal units $1 + 3^{c_0}\mathbb{Z}/3^c$ (hence $\equiv 1 \bmod 9$ for $c_0 \geq 2$). Each correction is a ratio of χ-lifts of the same lower-level element, which makes the property natural; for the first two descents it is proved (proposition 5, corrections are powers of $\rho = 1+3^{c-1}$); the general case is the parity-twisted recursion of proposition 5. The corrected chain of consequences, with the qualitative and quantitative branches kept separate, is:

carry depth ⇒ theorem 17 for all c ⇒ strict non-coboundary for every primitive sector ⇒ strict finite-level gap;

$$\text{short-witness coverage (Conjecture 5, strong)} \Rightarrow \inf_Q \kappa_Q > 0, \qquad \text{polylog coverage (weak)} \Rightarrow \kappa_Q \gg (\log Q)^{-2A}.$$

Generation alone (theorem 17 (i)) does *not* imply coverage: a fixed finite witness set is a fixed family of 3-adic units, and for $c \to \infty$ its coverage of all primitive characters is governed by the 3-adic digit patterns of finitely many logarithm ratios—a normality-type question. The realistic provable route to coverage is therefore *rapid generation*: witness families of slowly growing length (the weak form), where the comparison-graph/mode-spreading structure can act.

Controlled-length coverage audit

On the audited range the coverage statistic $\delta_c(L)$ is computed exactly:

c_0	c	$\#\mathcal{W}_c(1)$	$\delta_c(1)$	$\#\mathcal{W}_c(2)$	$\delta_c(2)$
2	3	7	1.9696	8	1.9696
2	4	10	1.4547	26	1.9966
3	5	75	1.9996	80	1.9996
4	6	241	1.99996	242	1.99996
5	7	728	1.999995	728	1.999995

Even loop-pair witnesses ($L = 1$) give near-maximal coverage, with one dip at $(c_0, c) = (2, 4)$ restored by $L = 2$. Caveat: on this diagonal c_0 grows with c; the decisive asymptotic regime (fixed c_0, $c \to \infty$) requires the parity-twisted recursion (Task A) to extend the multiplier tables, and is the next computation after it.

The missing ingredient, stated precisely

To convert proposition 4 into $\sup_c \gamma_c < 1$ by the comparison route one must propagate the approximate identities globally and contradict the exact orthogonality $\sum_{a \in \ker \chi_2} \chi(a) = 0$; by corollary 4 only the bounded-level core of this propagation is still needed, and Conjecture 5 is its sharpest form (the comparison graph below remains a useful route to it).

Definition 1 (Comparison graph)**.** Fix c and a branch bound B. The vertices are the states $U_{3^c} \times \mathbb{Z}/2$. There are two edge types. *Common-source edges*: for each state x and $b, b' \le B$ with $b \equiv b' \pmod 2$, join $S_b(x)$ to $S_{b'}(x)$ (these realise $v \sim 4^{-j}v$ at equal parity; $b' - b$ odd joins $(v, \sigma) \sim (2^{-t}v, \sigma + t)$ across parity). *Common-successor edges*: join $x = (a, \tau)$ to $x' = (a', \tau')$ if $S_b(x) = S_{b'}(x')$ for some $b, b' \le B$ (equivalently $3a' + 1 \equiv (3a + 1)2^{b-b'}$, which forces $b \equiv b' \bmod 2$ and $\tau' = \tau + (b - b')$). A path has *robust diameter* $\le D$ at mass threshold θ if any two vertices in the same $\ker \chi_2$-fiber are joined by a path of length $\le D$ all of whose edges use branches of weight $w_b \ge \theta$, and this remains true after deleting any edge set of stationary mass $< \theta$.

Conjecture 6 (Comparison-graph expansion; strong and weak forms)**.** (Strong.) *There are B, A such that for every c the comparison graph has robust diameter $O(c^A)$ at mass threshold $\gg c^{-A}$.* (Weak, sufficient for theorem 18.) *For every non-coboundary χ, from every high-mass state the approximate-coboundary phase propagates within $O(c^A)$ comparisons to representatives of a subgroup $H \le U_{3^c}$ with $\chi|_H \ne 1$ and $\left| |H|^{-1} \sum_{h \in H} \chi(h) \right| \le 1 - \eta$, $\eta \gg c^{-A}$. By corollary 4, both forms need only be established at a bounded level 3^{c_0}.*

Theorem 18 (Conditional polylog gap)**.** *Assume Conjecture 6. Then $1 - \gamma_c(s) \gg c^{-2A}$ uniformly for $s \in [0, 1]$; by theorem 4, $\kappa_{3^r}(s) \gg (\log Q)^{-2A}$, the quantitative form of the finite spectral hypothesis.*

Proof, modulo Conjecture 6. If $\gamma_c \geq 1-\varepsilon$, proposition 4 provides d satisfying the approximate coboundary identity off a defect set of mass $O(\varepsilon)$. By Conjecture 6 there is a comparison tree of depth $O(c^A)$ avoiding the defect set, along which the relation $\chi(a)d(x) \approx \chi(a')d(x')$ propagates with total accumulated error $O(D\varepsilon^{1/2})$, $D = O(c^A)$ the path length (Cauchy–Schwarz over the path). The $b, b{+}2$ comparisons make d approximately $\langle 4\rangle$-invariant on the tree, and Step 2 of theorem 3 then forces χ to be $O(D\varepsilon^{1/2})$-close to a constant on $\ker\chi_2$. Averaging, $\left|\sum_{a\in\ker\chi_2}\chi(a)\right| \geq \left(1-O(D\varepsilon^{1/2})\right)|\ker\chi_2|$, contradicting exact orthogonality (a non-coboundary χ is nontrivial on the squares) unless $D\varepsilon^{1/2} \gg 1$, i.e. $\varepsilon \gg D^{-2} = c^{-2A}$—the origin of the exponent $2A$. □

Remark (Why this and not monotonicity or single-orbit propagation). Monotonicity in c is false (part III), so no monotone conductor argument exists. Propagating $\langle 4\rangle$-invariance along the full $\langle 4\rangle$-orbit (length 3^{c-1}) costs a factor 3^c and yields only a polynomial-in-Q gap; lemma 4 shows the same 9^{-c} scale appears as the one-step damping of the least-damped primitive mode. The polylog/constant gap therefore needs short comparison paths (the expansion conjecture) or, equivalently, the spreading of $\eta \mapsto \eta\circ A$ across the character group. The numerical decay of γ_c beyond a fixed conductor is consistent with both mechanisms acting.

Block-cancellation alternative. The earlier Doeblin route (minorization in the sense of [24]) remains viable and is complementary: exhibit from every state a block family of untwisted mass $\geq \alpha$ whose endpoints contain a full coset of a subgroup H with $\chi|_H \neq 1$ (exact cancellation $\sum_{u\in H}\chi(uh) = 0$). For the killed strip, build the families with identical valuation multisets so the height weights match exactly, or prove bounded distortion $w_s(w_1)/w_s(w_2) \in [C^{-1}, C]$ with QSD bulk retention $O(e^{-cL})$ (quasi-stationary distributions of the killed operator; see [7]).

11 The keystone: bad orbits cast finite non-coboundary shadows

Throughout this paper the global implication has been kept strictly separate from the finite-operator theory. We now prove its first half: the *keystone arrow.* Call a Syracuse orbit *bad* if it is divergent or eventually enters a nontrivial cycle.

Lemma 26 (Drift)**.** *For every bad orbit,* $\limsup_N B_N/N \leq \log_2 3 + \log_2 \frac{7}{6} < 1.81 < 2$.

Proof. A non-cyclic orbit visits pairwise distinct integers, so $\sum_{n<N} 1/x_n = O(\log N)$, and $\log_2 x_N = \log_2 x_0 + N\log_2 3 - B_N + \sum_{n<N}\log_2(1+\frac{1}{3x_n})$ with $x_N \geq 1$ gives $B_N/N \leq \log_2 3 + O(\frac{\log N}{N})$. On a nontrivial cycle of period p, $2^{B_p} = 3^p\prod(1+\frac{1}{3x_i})$ and $x_i \geq 2$ give $B_p/p \leq \log_2 3 + \log_2\frac{7}{6}$. (The stationary mean is $\mathbb{E}_{\mathrm{geom}}[b] = 2$, attained with equality only by the trivial cycle, where $b \equiv 2$.) □

Remark (Calibration: the popular "ratio", the drift line, and a terminology dictionary). It is worth recording how the drift coordinate B_N/N meets the heuristic vocabulary of the amateur and survey literature, where a single trajectory statistic appears under many incompatible names. The folklore "average sequence" argument (each odd step multiplies by ≈ 3 and the ensuing run of halvings divides by $\sum_{k\geq 1} k2^{-k} = 2$ on average, for a net factor 3/4) is exactly the stationary law $\mathbb{E}_{\mathrm{geom}}[b] = 2$ used in lemma 26: a number x_0 is *expected* to take $\log_2 x_0/(2-\log_2 3)$ odd steps to reach 1. Writing $N(x_0)$ for the actual odd-step count, the popular *length ratio* $\rho(x_0) := N(x_0)\,(2-\log_2 3)/\log_2 x_0$ is then *not* an independent observable: with $\bar{b}_N := B_N/N$ the mean letter and $\delta_N := \sum_{n<N}\log_2(1+\frac{1}{3x_n})$ the *same* correction that appears in lemma 26, the descent identity $\log_2 x_0 = N(\bar{b}_N - \log_2 3) - \delta_N$

gives

$$\rho(x_0) \;=\; \frac{2-\log_2 3}{\bar b_N - \log_2 3 - \delta_N/N}, \qquad \delta_N/N \to 0.$$

Thus the "ratio" is just a reparametrised drift: ρ is large *precisely* when the orbit's mean drift $\bar b_N$ hovers just above the critical equidistribution line $\log_2 3$, and the recurrent amateur conjecture that ρ is *unbounded* is exactly the statement that record orbits drive $\bar b_N \downarrow \log_2 3$—i.e. that they approach, but by lemma 26 never cross, the deterministic-equidistribution barrier this paper localises. (Numerically the record-setters $27, 230631, 63728127, 166763117679$ give $\rho = 3.58, 3.82, 5.72, 5.24$ against the identity values $3.39, 3.77, 5.66, 5.22$, the gap being δ_N/N, which decays with N.) Two cautions follow. First, these records are *not* biased-alphabet objects: their measured binary $\{1,2\}$-bias stays $\lesssim 0.1$, far below the alphabet-capacity ceiling $\varepsilon_{\rm bin}$ of the companion phase diagram—a champion rides the natural measure for an anomalously long time rather than conspiring inside a sub-alphabet. Second, the empirical ceiling of ρ near 5.7 is a *count* ratio and must not be conflated with the band-magnitude capacity $2^{2D^*} \approx 5.63$ of the narrow-band analysis; the per-orbit excursion $\max_N(B_N - N\log_2 3)$ grows without bound, and the numerical proximity of the two constants is a coincidence.

The naming is genuinely balkanised; we fix the correspondences once.

Literature term (source)	**Object here**
total stopping time σ_∞ [20, 21]	steps to reach 1 : $N + B_N$
stopping time σ, "glide" [37, 28]	first N with $B_N > N\log_2 3$ (drift <0)
parity vector / encoding [37, 12]	the word (b_n) (= 2-adic coding [6])
Syracuse / accelerated / odd map	the odd orbit (x_n)
"shortcut" $T(n) = (3n+1)/2$	one letter, aggregated over a run
"drop" / descent	the letter $b_n = v_2(3x_n+1)$
expansion / max value / "completeness" [28]	excursion $\max_N(B_N - N\log_2 3)$
length "ratio" / delay records [28]	$\rho = (2-\log_2 3)/(\bar b_N - \log_2 3)$ (above)

Every popular "how long is the orbit" record is, in these coordinates, a record of how closely $\bar b_N$ approaches $\log_2 3$ from above; the content of the paper is that this approach is finitely shadowed.

Lemma 27 (Letter characters are two-point shadows)**.** *For any character χ' modulo 3^r, the finite-level two-point function*

$$F_{\chi'}(x_n, x_{n+1}) := \chi'(x_{n+1})\,\overline{\chi'}(3x_n+1) = \chi'(2)^{-b_n}$$

is exactly a character of the letter b_n, of order equal to $\operatorname{ord}\chi'$ *(since 2 generates the units). By the classification* $\mathrm{Cob} = \langle\chi_2\rangle$ *(theorem 3), the letter characters realized by* non-coboundary χ' *are exactly those with nontrivial* 3*-part; the exempt directions are the trivial character and the parity* $(-1)^{b_n}$ *(realized by χ_2, the valuation-parity identity).*

Theorem 19 (Keystone arrow)**.** *Every bad orbit casts a finite non-coboundary spectral shadow: there exist r and a non-coboundary character χ' modulo 3^r such that the empirical averages $\frac{1}{N}\sum_{n<N}\chi'(x_{n+1})\overline{\chi'}(3x_n+1)$ do* not *converge to the stationary value $\mathbb{E}_{\rm geom}\big[\chi'(2)^{-b}\big]$.*

Proof. Suppose not: every such average converges to its stationary value. By lemma 27 this says the empirical letter averages $\frac{1}{N}\sum \xi(b_n)$ converge to $\mathbb{E}_{\rm geom}[\xi(b)]$ for every letter character ξ of modulus

$2 \cdot 3^j$, $j \geq 1$, *except possibly* $\xi =$ parity. Pass to a subsequence realizing $\beta := \liminf B_N/N \leq 1.81$ (lemma 26) and let ν be the corresponding empirical limit law of the letter b. Truncation gives $\mathbb{E}_\nu[b \wedge K] \leq \beta$ for all K, hence tightness. The law of b modulo $2 \cdot 3^j$ under ν matches the geometric law in every character except parity, so

$$\Pr_\nu[b \equiv k\ (2 \cdot 3^j)] = \Pr_{\text{geom}}[b \equiv k\ (2 \cdot 3^j)] + \frac{t\,(-1)^k}{2 \cdot 3^j} \quad \text{for some } |t| \leq 2:$$

the single deviating direction *spreads* over the residues, with pointwise correction $O(3^{-j})$. Letting $j \to \infty$ and using tightness, every point mass matches exactly: $\Pr_\nu[b = k] = 2^{-k}$ for all k. These sum to 1 (no escaping mass), so $\mathbb{E}_\nu[b] = 2$—contradicting $\mathbb{E}_\nu[b] \leq \beta < 2$. □

Proposition 15 (Drift visibility at the tilted trivial character)**.** *For $s \in (0,1)$, the trivial-character defect average of a bad orbit satisfies, by Jensen and lemma 26,*

$$\frac{1}{N}\sum_{n<N} 2^{\,s(\log_2 3 - b_n)} \;\geq\; 2^{\,s(\log_2 3 - B_N/N)} \;\xrightarrow[N]{}\geq\; 2^{\,s(\log_2 3 - 1.81)} > 1 > \frac{3^s}{2^{1+s}-1} = \mathbb{E}_{\text{geom}}\big[2^{s(\log_2 3 - b)}\big]:$$

the s-tilt alone already separates bad orbits from stationarity, with equality of the two endpoints exactly at $s \in \{0,1\}$ (the Collatz criticality $3 = 2^{1+1} - 1$).

Remark (The arrow is one-directional). The converse of theorem 19 fails, and this matters for the correct formulation of what remains. The *trivial cycle* itself ($x = 1$, $b \equiv 2$) has empirical shadow value $\chi'(2)^{-2}$, which differs from $\mathbb{E}_{\text{geom}}[\chi'(2)^{-b}]$ for general non-coboundary χ'; hence *every convergent orbit* also casts a non-stationary shadow (its averages converge to the trivial-cycle value, not the geometric one). A naive "no orbit deviates" rigidity statement is therefore *false* if the Collatz conjecture is true. The correct remaining statement is an exclusion:

Open Problem 1 (Quenched shadow exclusion)**.** *No positive Syracuse orbit that is* not eventually trivial *supports a persistent non-coboundary two-point shadow deviation from the geometric stationary value. Equivalently: for every orbit not absorbed by* 1*, every finite non-coboundary shadow either converges to the stationary value or forces descent within bounded time. Combined with theorem 19, this excludes bad orbits and implies the full Collatz conjecture. Two cautions:* (i) *if Collatz is true the statement is vacuous on positive integers, so it cannot be tested on known convergent orbits;* (ii) *it is essentially the Collatz conjecture in minimal finite-shadow language—the value of the reduction is the finiteness and explicitness of the obstructing object, not a decrease in logical strength.*

Possible routes. Three attack shapes, all open and all at the deterministic-equidistribution barrier. *(1) Finite inverse theorem for shadow bias:* persistent deviation $\geq \varepsilon$ forces, via a density-increment argument and the sliding-window congruence, a recurrent structured residue trap modulo 3^R; prove any such recurrent trap is an exact algebraic obstruction, hence coboundary by theorem 3, hence impossible for non-coboundary χ'. *(2) Shadow deviation forces descent:* for every (r, χ', ε) there are (M, H) such that a deviation $\geq \varepsilon$ over a length-M window forces $x_{n+h} < x_n$ within a bounded enlargement—local and dynamical, avoiding global normality: the most realistic target. *(3) Empirical-measure rigidity:* every non-eventually-trivial positive-orbit limit measure has annealed shadow marginals—a measure classification of exactly the type that is the known wall (symbolic models admit many invariant measures; the integer-orbit constraint is the arithmetic heart).

Three finite approaches to quenched exclusion

Theorem 20 (Periodic words cast shadows)**.** *Every positive periodic Syracuse orbit other than the trivial cycle casts a finite non-coboundary shadow: some non-coboundary χ' has cycle average $\frac{1}{p}\sum_i \chi'(2)^{-b_i} \neq \mathbb{E}_{\text{geom}}[\chi'(2)^{-b}]$.*

Proof. The cycle's letter law is a finitely supported probability measure. If it matched the geometric law in every letter character except parity, the spreading argument of theorem 19 (the parity correction is $O(3^{-j})$ pointwise at modulus $2 \cdot 3^j$) would force $\Pr[b = k] = 2^{-k}$ for *every* k—impossible for finite support. The detecting character has order with nontrivial 3-part, hence is non-coboundary by theorem 3. (The trivial cycle also deviates, $b \equiv 2$; it is the designated exception throughout.) □

Corollary 6. *Cycle elimination is a special case of quenched shadow exclusion (open Problem 1). Computationally: all integer cycles with period $p \leq 7$ are rotations of the trivial cycle (exhaustive search over $2^B > 3^p$ with $(2^B - 3^p) \mid A_w$), and every sampled nontrivial periodic word is detected already at character order 3 or 9.*

Proposition 16 (Finite inverse theorem: elementary layers)**.** *Let $z = \chi'(2)$, $m = \operatorname{ord} z$, and $D_N(\chi') = \frac{1}{N}\sum_{n<N} z^{-b_n} - \mathbb{E}_{\text{geom}}[z^{-b}]$. If $|D_N| \geq \varepsilon$ then:* (i) *some residue class $a \bmod m$ of the letters is biased, $\left|\frac{1}{N}\#\{n : b_n \equiv a\} - \Pr_{\text{geom}}[b \equiv a]\right| \geq \varepsilon/m$ (Fourier inversion on $\mathbb{Z}/m$);* (ii) *equivalently, by $F_{\chi'} = \chi'(x_{n+1})\overline{\chi'}(3x_n+1)$, the empirical* edge *measure on $U_{3^r} \times U_{3^r}$ deviates from the stationary edge law by $\gg_r \varepsilon$ on an explicit residue-edge set:* shadow bias is a finite residue-edge trap*.* *The third layer*—persistent trap ⇒ descent or coboundary obstruction*, with the latter impossible for non-coboundary χ' by theorem 3—is the open target; we emphasize, per the warning above, that it would constitute a genuinely new deterministic rigidity principle, not a corollary of the finite gap.*

Proposition 17 (C1: mean-visible deviation forces descent)**.** *If a window $[N, N+M)$ of a positive orbit has mean letter $\frac{1}{M}\sum b_n \geq \log_2 3 + \eta$ with $\eta > \left(3x_N \ln 2\right)^{-1}$, then $\min_{N<n\leq N+M} x_n < x_N$: descent within the window.*

Proof. If $x_n \geq x_N$ throughout, then $\log_2 x_{N+M} = \log_2 x_N + M\log_2 3 - \sum b_n + \sum \log_2(1 + \frac{1}{3x_n}) \leq \log_2 x_N - \eta M + \frac{M}{3x_N \ln 2} < \log_2 x_N$, a contradiction. □

The C1/C2 split, empirically. A shadow deviation therefore splits: *mean-visible* deviations (letter mean displaced) force descent at once by proposition 17; *phase-only* deviations (mean near critical, non-coboundary phase biased) are the true remaining obstruction. A window audit over long convergent orbits (648 windows of length 64, starts $\sim 10^{80}$) shows the split is real and balanced: the fraction of windows with displaced mean rises from 4% at low deviation to 53% at high deviation, and phase-only high-deviation windows exist with letter mean throughout $[1.78, 2.25]$. (On convergent orbits every window descends, so the positive direction of "deviation forces descent" is untestable there—exactly the vacuousness caution of open Problem 1; the informative content is the split structure.) The sharpest remaining route is thus:

phase-only shadow bias ⇒ residue-edge trap ⇒ coherent lift attempt ⇒ coboundary (excluded) or descent.

The last arrow: max-plus certificates and where C2 actually lives

Fix r, a non-coboundary χ', $z = \chi'(2)$, a phase ω, and $\varepsilon > 0$. On the residue-edge graph G_r (vertices U_{3^r}, edges $a \xrightarrow{b} (3a+1)2^{-b}$) put the two labels

$$g(b) = \log_2 3 - b, \qquad \phi(b) = \Re\big(\omega\,(z^{-b} - \mathbb{E}_{\text{geom}}[z^{-b}])\big).$$

Lemma 28 (Certificate forces descent)**.** *Let $H \subseteq G_r$ contain all edges traversed by an orbit window $[N, N+M)$ with phase bias $\frac{1}{M}\sum \phi(b_n) \geq \varepsilon$. Suppose there exist $\Psi : U_{3^r} \to \mathbb{R}$, $\lambda \geq 0$, $\delta > 0$ with*

$$g(b) + \lambda\big(\phi(b) - \varepsilon\big) + \Psi(a') - \Psi(a) \leq -\delta \qquad \textit{on every edge of } H.$$

Then for $M > \operatorname{osc}(\Psi)/\big(\delta - (3x_N \ln 2)^{-1}\big)$ the orbit descends within the window. Dually (max-plus), such a certificate exists iff $\inf_{\lambda\geq 0}\big(P_H(\lambda) - \lambda\varepsilon\big) < 0$, where $P_H(\lambda)$ is the maximum directed-cycle mean of $g + \lambda\phi$ on H.

Proof. Telescoping the certificate along the window and using $\sum(\phi-\varepsilon) \geq 0$: $\sum g(b_n) \leq -\delta M + \operatorname{osc}\Psi$. With $\log_2 x_{N+M} = \log_2 x_N + \sum g(b_n) + \sum \log_2(1 + \frac{1}{3x_n})$ and the no-descent assumption $x_n \geq x_N$, the right side drops below $\log_2 x_N$—contradiction. The duality is the standard max-plus (Karp [18]) characterization of the maximum cycle mean. $\square$

Proposition 18 (The 3-adic-only formulation of C2 is false)**.** *Certificate existence cannot hold at any fixed 3-adic level, nor in the 3-adic tower alone:* (i) *the all-ones trap ($b \equiv 1$; $g \equiv \log_2 3 - 1 > 0$, constant phase) is coherent modulo 3^R for every R, biased, descent-free, and admits no certificate at any level ($P_H(\lambda) - \lambda\varepsilon \equiv \log_2 3 - 1$); it is eliminated* only *by the 2-adic tower: $b_n \equiv 1$ forever forces $x \equiv -1 \bmod 2^k$ for all k, i.e. $x = -1$ (the famous -1 cycle [20]).* (ii) *the critical $\{1,2\}$-mixture trap (letter mean $\log_2 3$, $\Pr[b{=}1] = 2 - \log_2 3$) is 2-adically frequency-feasible yet uncertified at $r = 2$ (margin $+0.056$).* (iii) *of the observed phase-only high-deviation windows on convergent orbits, 7 of 8 are uncertified at $r = 2$: observed traps contain spurious cycles, so any true statement must be projective, not fixed-level.*

Open Problem 2 (C2, final form: joint-tower max-plus rigidity)**.** *Every $(2,3)$-adically coherent, non-coboundary–biased, descent-free trap—a projective family of residue-edge traps over the* mixed *moduli 2^a3^b with nonvanishing mass, bias $\geq \varepsilon$, and near-critical drift—admits a max-plus descent certificate at some level of the joint tower. By lemma 28 this implies the quenched shadow exclusion (open Problem 1), hence the Collatz conjecture. The 2-adic constraints are not optional: by proposition 18 they supply the negative weight that the 3-adic graph alone cannot see. This is where the present program rejoins its predecessor: the mixed-modulus (2^r3^s) certificates of the stabilized-residue program [17] are exactly finite verifications in this joint tower.*

The final route, corrected once more:

$$\text{phase-only bias} \Rightarrow \text{joint-tower trap} \Rightarrow \begin{cases} \text{max-plus certificate} \Rightarrow \text{descent (lemma 28)}, \\ \text{no certificate at any level} \Rightarrow \text{a coherent } (2,3)\text{-adic object}, \end{cases}$$

and the remaining theorem is that the only such objects are the classified coboundary/trivial ones. C2 in full remains open; what is proved here is the mechanism (lemma 28), the failure of every purely 3-adic formulation (proposition 18), and the exact location of the remaining content.

Remark (Counterexample profile). A counterexample to the Collatz conjecture, if it exists, must by the present theorem stack be *mean-critical* ($\frac{1}{N}\sum b_n = \log_2 3 + o(1)$, else proposition 17), *non-coboundary phase-biased* (theorem 19), jointly $(2,3)$-adically coherent, and certificate-free at every level of the mixed tower. The simplest symbolic models with these properties are balanced low-letter words over $\{1,2\}$ with $\Pr[b{=}1] = 2-\log_2 3 \approx 0.415$ (Sturmian/Beatty type [25, 22], slope $2-\log_2 3$)—the critical analogues of the all-ones word, whose 2-adic point is $x = -1$. Direct realizability tests support the ghost expectation: for twelve intercepts θ and prefixes of length 800 ($B_N = 1268$), the unique residue class modulo 2^{B_N+1} realizing the Sturmian prefix has least positive representative of full bit-length $\approx B_N$, wandering with the prefix and never stabilizing—the signature of a mixed-adic ghost, not a positive integer. For *periodic* balanced traps the ghost exclusion is precisely the classical cycle problem, where the archimedean negative weight is supplied by linear forms in logarithms ($|2^B - 3^p| \gg 2^B/B^C$, [2, 27]): joint-tower rigidity restricted to periodic traps is thus partially within reach of existing transcendence technology, and the genuinely new content of open Problem 2 is its aperiodic, single-orbit analogue.

Calibrated measures, and the periodic case via linear forms in logarithms

Lemma 29 (Calibrated measures)**.** *Suppose a coherent mixed-tower trap with non-coboundary bias $\geq \varepsilon$ admits no max-plus certificate at any level. Then there exists a shift-invariant probability measure μ on the valuation sequence space, with adelically coherent residue marginals, such that*

$$\mathbb{E}_\mu[b] \leq \log_2 3, \qquad \int \phi \, d\mu \geq \varepsilon.$$

Consequently $\Pr_\mu[b{=}1] \geq 2 - \log_2 3 \approx 0.415$, *and if moreover* $\mathbb{E}_\mu[b] = \log_2 3$ *and μ is ergodic, the discrepancy cocycle $S_j = j \log_2 3 - B_j$ is recurrent μ-a.e. (Atkinson [1]). Thus open Problem 2 is equivalent to:* no such calibrated measure is realized by a positive integer orbit.

Proof. At each level $L = 2^a 3^b$, no certificate means $\inf_{\lambda \geq 0}\big(P_{H_L}(\lambda) - \lambda\varepsilon\big) \geq 0$. The cycle-mean vectors $(\overline{g}, \overline{\phi})$ of the trap span a compact convex polytope C_L, and by Sion's minimax theorem [34],

$$\inf_{\lambda \geq 0} \max_{\mu \in C_L} \big[\langle \mu, g\rangle + \lambda(\langle \mu, \phi\rangle - \varepsilon)\big] = \max\big\{\langle \mu, g\rangle : \ \mu \in C_L, \ \langle \mu, \phi\rangle \geq \varepsilon\big\},$$

so no certificate yields an invariant edge measure μ_L with $\int g \, d\mu_L \geq 0$ and $\int \phi \, d\mu_L \geq \varepsilon$. The condition $\int g \geq 0$ gives $\mathbb{E}[b] \leq \log_2 3$ uniformly, hence tightness of the letter marginals; a diagonal weak-$*$ limit along the tower produces a consistent projective family, i.e. a shift-invariant measure on the inverse limit with the same inequalities. The letter bound follows from $\mathbb{E}[b] \geq 2 - \Pr[b{=}1]$; recurrence at exact calibration is Atkinson's theorem [1] for zero-mean ergodic cocycles. $\square$

Theorem 21 (Periodic exclusion via linear forms in logarithms)**.** *There are effective constants C, τ (one may take $\tau = 13.3$, by Rhin's bound [27] $|B \ln 2 - p \ln 3| \geq c\, B^{-\tau}$) such that every nontrivial positive integer cycle of period p satisfies*

$$x_{\min} \leq C\, p^{\tau+1}.$$

Hence, given the verified range $x_{\min} > 2^{71}$ [4], every nontrivial cycle has $p > (2^{71}/C)^{1/(\tau+1)}$; and for every fixed period range the balanced (calibrated) periodic traps are effectively excluded: the linear form in logarithms [2] is the max-plus descent certificate for periodic traps. (Cycle bounds of this shape go back to Steiner [35], with the sharpest published forms in Eliahou [11] and Simons–de Weger [32].)

Proof. For a cycle, $2^B = 3^p \prod_i (1 + \frac{1}{3x_i})$, so $\theta := B - p\log_2 3 = \log_2 \prod_i (1 + \frac{1}{3x_i}) \in \big(0,\, p/(3x_{\min}\ln 2)\big]$; positivity and integrality pin $B = \lceil p\log_2 3\rceil$, so $B \le 2p$. Rhin's effective bound [27] gives $\theta \ge c'B^{-\tau} \ge c''p^{-\tau}$. Combining, $c''p^{-\tau} \le p/(3x_{\min}\ln 2)$, i.e. $x_{\min} \le Cp^{\tau+1}$. $\square$

Corollary 7 (Where a counterexample must live). *Combining lemma 29 and theorem 21: if open Problem 2 fails, the witnessing calibrated measure is* aperiodic*, supported on low-letter sequences with* $\Pr[b{=}1] \ge 0.415$*, with recurrent discrepancy and non-coboundary bias—the Sturmian ghost class of section 11—and all of its periodic approximants of any fixed period range are effectively excluded by theorem 21. The remaining content of the program is thus the aperiodic, single-orbit analogue of the linear-forms certificate:* Baker along orbits*.*

Repetition rigidity: exclusion of low-complexity calibrated words

The aperiodic analogue can now be carried out, and it closes more strongly than anticipated: the contraction bookkeeping of the repeated-block heuristic is unnecessary, because 2-adic realizability gives *exact* rigidity. Recall the classical coding [37, 12, 6]: for a valuation block $u = (b_0, \dots, b_{p-1})$ with sum B_u, the set of odd x whose first p letters equal u is a *single residue class modulo* 2^{B_u+1} (machine-checked: 84,471 instances, no exceptions), and $B_u \ge p$.

Theorem 22 (Repetition rigidity). *Let $(x_n)_{N\le n\le N+M}$ be a positive Syracuse orbit segment with $x_n \in [Y, WY]$ throughout, and let p satisfy $2^{p+1} > Y(W-1)$. If the segment's valuation word has fewer than $M-p+1$ distinct factors of length p, then $x_n = x_m$ for some $N \le n < m \le N+M-p$: the orbit is an integer cycle of period $\le M$ and minimum $\ge Y$.*

Proof. By pigeonhole two positions $n < m$ carry the same length-p block u; by the coding, $2^{B_u+1} \mid x_m - x_n$ with $2^{B_u+1} \ge 2^{p+1} > Y(W-1) \ge |x_m - x_n|$; hence $x_m = x_n$, and the orbit repeats. $\square$

Corollary 8 (Bounded-discrepancy aperiodic exclusion). *No aperiodic valuation word with bounded discrepancy* $\sup_N |N\log_2 3 - B_N| \le D$ *and subexponential factor complexity ($P(p) \le 2^{\,p-2D-c_0}$ for large p) is realizable by a positive Syracuse orbit. In particular* no Sturmian word of any slope *($D < 1$, $P(p) = p{+}1$; [25, 22]) is realizable: the critical ghost class of section 11 is excluded unconditionally—no linear-forms input is needed for the qualitative statement, since the manufactured cycle's word is periodic, contradicting aperiodicity.*

Proof. A realization from x_0 is confined to the band $[x_0 2^{-D-1}, x_0 2^{D+1}]$ for $M \le x_0$ steps (the $\log_2(1+\frac{1}{3x})$ corrections total $o(1)$). Take $p = \lceil\log_2 x_0\rceil + 2D + 3$ and $M = p + P(p) + 1 \le x_0$ (subexponential complexity; small x_0 is covered by the verified convergence range [4]). theorem 22 produces a cycle, whose word is eventually periodic—contradiction. $\square$

Theorem 23 (Complexity–height tradeoff). *With the effective constants of theorem 21: every positive orbit segment of length M confined to a dyadic band $[Y, 2Y]$ with $Y > CM^{\tau+1}$ has word complexity*

$$P(p) \;\ge\; M - p + 1, \qquad p = \lceil \log_2 Y\rceil + 2 :$$

a no-descent stretch at height Y must exhibit nearly full factor diversity *at scale $\log_2 Y$. Equivalently, any banded no-descent window longer than $(Y/C)^{1/(\tau+1)}$ forces the word to mimic randomness at that scale.*

Proof. Otherwise theorem 22 manufactures a cycle with period $\le M$ and minimum $\ge Y$, and theorem 21 gives $Y \le CM^{\tau+1}$—contradiction. □

Corollary 9 (The ghost class is closed; the frontier moves)**.** *Combining lemma 29 and theorem 23: a counterexample orbit whose discrepancy path admits banded windows of unbounded length at heights Y_k (the behaviour forced at the measure level by Atkinson recurrence) must carry a word of factor complexity at least $(Y_k/C)^{1/(\tau+1)} - p_k$ at scales $p_k \approx \log_2 Y_k$:* exponential rate $\ge 1/(\tau+1)$ per bit*. All low-complexity ghosts—Sturmian, morphic, linearly recurrent, polynomial, and everything below that exponential rate—are excluded. The Aperiodic Baker Rigidity target of the route map is thereby proved in a strengthened form, with the repeated block replacing the closed cycle exactly as envisaged, but with exact* 2*-adic equality in place of the contraction squeeze. What survives is precisely: orbits whose valuation words have near-full exponential complexity inside every banded window (randomness-mimicking), or whose discrepancy path forever avoids long banded windows; for these the S-unit/subspace frontier is the next tool. We are not aware of prior statements of theorems 22 and 23 and corollary 8, though their ingredients (2-adic coding [37, 12, 6]; cycle bounds via linear forms [35, 11, 32]) are classical.*

Remark (The wall, in its sharpest form). open Problem 1 is the quenched-equidistribution wall of the earlier program, now reduced to its minimal finite form: single-orbit Birkhoff averages of finite-level functions. The finite operator theorem of this paper ($\sup_c \gamma_c(0) < 1$) is its exact *annealed* counterpart: the transfer operator mixes these very characters at a uniform exponential rate. What is missing is the passage from annealed mixing to pointwise convergence along the deterministic forward orbit—the step that no current technique supplies for a non-invertible arithmetic map with no invariant measure on the integers. The contribution of the keystone arrow is to make the two sides meet at the same finite object: one two-point character family, with the coboundary direction classified, exempted, and shown harmless.

Conclusion

The defect cocycle relocates rather than removes the difficulty, but the finite-operator side is now substantially a theorem. The rigorous core comprises the sliding-window congruence; the proof that the quadratic character is valuation parity in disguise, hence an exact coboundary; the *classification* $\mathrm{Cob} = \langle \chi_2 \rangle$ (theorem 3), which makes the non-coboundary quotient canonical rather than conjectural; the strict spectral gap at every finite level (corollary 2); the conductor collapse (theorem 4), which reduces uniformity in Q to the single computable sequence $\gamma_c(s)$; the height-strip rotation identity (proposition 3); the conductor-raising ejection theorem (theorem 8); and the cascade equivalence, mass bound, two-point profile, and deeper-channel bound (theorem 9, lemmas 12 and 13, and proposition 12). The finite-operator endpoint narrowed in stages, each closed by an honest negative finding: descent (theorems 5 and 6) makes the three deepest digits lossless but *stalls there* (proposition 6); every counting and trace assembly terminates at exact marginality (proposition 10); and the coarse Schur regimes fail precisely where the degenerate class decouples from endpoint matching. What survives is the threshold $m = c$: the complete-kernel formula (proposition 11) and the adjoint-side Schur form reduce the uniform gap Conjecture 1 to the threshold Schur bound (Conjecture 4). Within it, the cascade (class-II/III) budget is now *proved contractive* (lemma 14 and proposition 13: sharp pinning constant $C_0 < 1.355$, heavy-row rate $(3/4)^c$, all-row rate $(\sqrt{3}/2)^c$) modulo a single analytic input—threshold flattening, reduced in theorem 10 to an exact, character-free renewal Fourier-decay statement—with the class-IV budget subordinate by audit. The flattening input is *proved* (theorem 10: bounded shells, $\tilde{Q}_L \le 3C_1 < 5.16$, by the digit-extension energy argument), so the cascade sector of the threshold Schur bound is an unconditional theorem (theorem 14); and the full finite H23a operator theorem $\sup_c \gamma_c(0) < 1$ is now *proved* (theorem 13 and corollary 5), with the class-IV budget closed by the root-ball analysis (lemmas 22 to 25 and theorem 12) at the cascade-sector rate $(\sqrt{3}/2)^c$ and asymptotic gap constant $1 - (\sqrt{3}/2)^{1/2} \approx 0.069$. The probabilistic side is closed—nominal-shell mass, leading-coefficient structure, singleton rigidity, and the level-cost law itself are all theorems (lemmas 17 and 18 and theorem 11), the last with the exact constant $\lambda = \frac{4}{7} < 3^{-1/2}$ via the parity-trichotomy and mod-9 orbit-spread argument. Small c is covered by the strict gap (corollary 2) and finiteness—no certified numerics needed for the qualitative statement. The earlier conditional routes—comparison-graph expansion (Conjecture 6), short-witness coverage (Conjecture 5), the bounded-level holonomy audit, and the canonical witness (theorem 17) as a route to uniformity—are retained as strictness, diagnostic, and conditional results, but are superseded as routes to the uniform gap. The heavy-row audits support contraction by mass-plus-profile bookkeeping alone (per-level decay ≈ 0.72; no signed cancellation available or required), and the deeper-channel saving is proved. To be precise about scope: it is the *finite H23a operator theorem* that is proved here—and the keystone arrow (bad orbit $\Rightarrow$ finite non-coboundary spectral shadow) is now also proved (theorem 19), with the coboundary classification supplying exactly the exempt direction. The converse of the arrow fails (the trivial cycle itself deviates, section 11), so the remaining statement is the *quenched shadow exclusion* (open Problem 1): no non-eventually-trivial orbit may sustain such a deviation. That exclusion implies the Collatz conjecture; it is essentially the conjecture in minimal finite-shadow language, and it sits exactly at the deterministic-equidistribution barrier, with three identified attack routes of which "shadow deviation forces descent" is the most realistic.

Lean formalization. The elementary core of the repetition rigidity argument is machine-verified in Lean 4 [9] (Mathlib [23] v4.27.0): `CollatzLean/FiniteSpectralShadows.lean` proves `word_congr` (the 2-adic coding lemma), `repetition_rigidity`, and `orbit_periodic_of_eq` sorry-free, depending only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`); the Baker cycle bound is stated as the leading `proof_wanted` of the formal program. Full provenance for every computation and proof step in this paper is recorded in the project's AIPM (`ailog`) trail.

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